CHAPTER 10

Nitrogen- and Phosphorus-Containing Compounds

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 965e99720454ecb4108d9722e7f1dd9d26005a1329ce0de844da2e161de370e1

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch10:exercise:glycine-charge | glycine-charge:stage-1, glycine-charge:stage-2, glycine-charge:stage-3

ch10:exercise:peptide-link | peptide-link:stage-1, peptide-link:stage-2, peptide-link:stage-3

ch10:exercise:strecker-map | strecker-map:stage-1, strecker-map:stage-2, strecker-map:stage-3

ch10:exercise:phosphate-links | phosphate-links:stage-1, phosphate-links:stage-2, phosphate-links:stage-3

ch10:lesson:amino-acids | amino-acids:rule, amino-acids:cue, amino-acids:trap, amino-acids:app, amino-acids:step-1, amino-acids:step-2, amino-acids:step-3

ch10:lesson:peptides | peptides:rule, peptides:cue, peptides:trap, peptides:app, peptides:step-1, peptides:step-2

ch10:lesson:synthesis | synthesis:rule, synthesis:cue, synthesis:trap, synthesis:app, synthesis:step-1, synthesis:step-2, synthesis:step-3

ch10:lesson:phosphorus | phosphorus:rule, phosphorus:cue, phosphorus:trap, phosphorus:app, phosphorus:step-1, phosphorus:step-2

ch10:molecule:alanine | alanine

ch10:molecule:glycine | glycine

ch10:molecule:glycylglycine | glycylglycine

ch10:molecule:aminonitrile | aminonitrile

ch10:molecule:dimethylphosphate | dimethylphosphate

ch10:molecule:phosphoricacid | phosphoricacid

ch10:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 10.1 Amino Acids, Peptides, and Proteins

scope-2: 10.2 Synthesis of α-Amino Acids

scope-3: 10.3 Phosphorus-Containing Compounds

Objective: Analyze amino-acid charge and stereochemistry

Objective: Identify peptide bonds and synthesis carbon sources

Objective: Distinguish phosphate linkages and reaction energetics

Supplied Kaplan curriculum: Chapter 10; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter10b.xhtml

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

amino-acids | Amino acids, charge, and stereochemistry

Ionization depends on pH

amino-acids:rule | Core Rules

An alpha amino acid has amino and carboxyl groups attached to the same alpha carbon. At many near-neutral pH conditions, simple amino acids predominantly form zwitterions. Side chains add nonpolar, aromatic, polar, acidic, or basic behavior. Glycine has no stereocenter; most proteinogenic L amino acids are S, with cysteine the familiar R exception.

amino-acids:cue | Memory Cues

Low pH adds protons; high pH removes them. Net charge is the sum of all ionized groups.

amino-acids:trap | Common Mistakes / MCAT Traps

A zwitterion has formal charges despite net zero charge. The pI is the pH of average net zero charge; for ionizable side chains, average the two pKa values bracketing the neutral species, not automatically the terminal pKa pair.

amino-acids:app | MCAT Applications

Aliphatic side chains such as Ala, Val, Leu, and Ile are nonpolar; Ser and Thr have polar OH groups; Asn and Gln have polar amides. Asp/Glu are usually negatively charged near neutral pH; Lys/Arg usually positive. Histidine protonation is strongly pH- and environment-dependent. Aromatic does not always mean purely nonpolar: Tyr has a phenolic OH. At pH below pI a protein tends positive; above pI it tends negative.

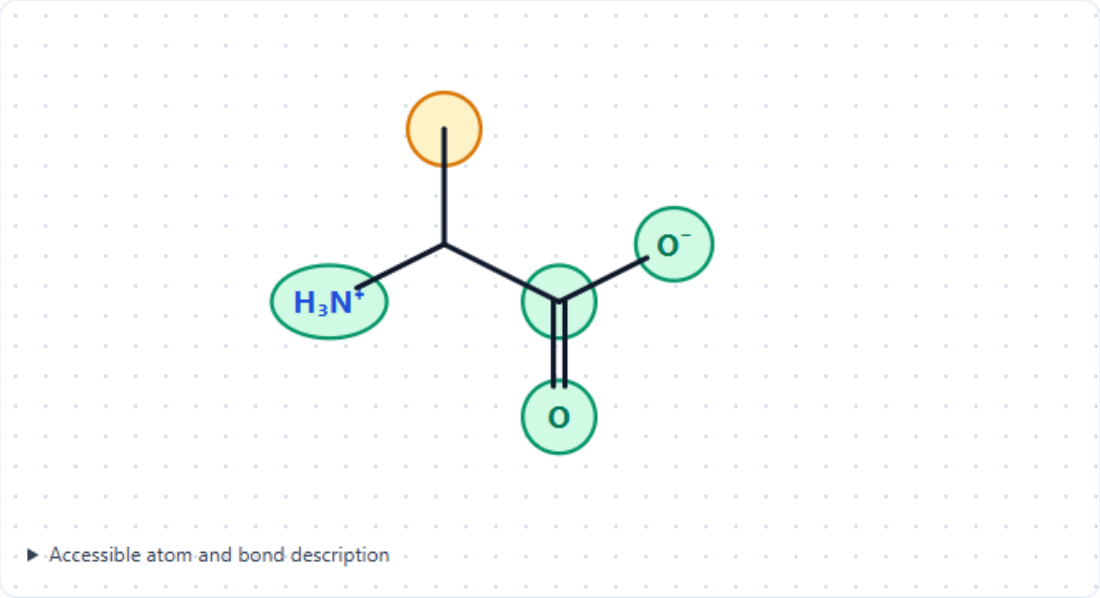
amino-acids:step-1 | 1. Zwitterionic alanine

The nitrogen bears positive charge and the carboxylate bears negative charge. Its alpha carbon also has methyl and H. This connectivity drawing leaves stereochemistry unspecified.

Displayed molecule ID: alanine

{"highlights": {"prin": [0, 2, 3, 4], "sub": [5]}, "revealName": true}

Screenshot: amino-acids-step-1



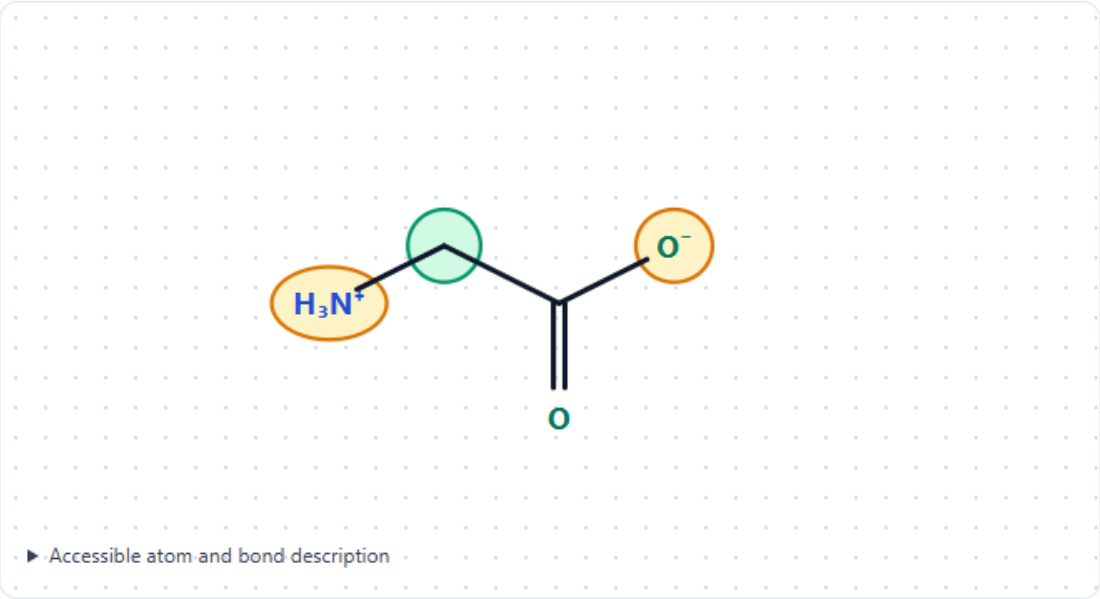
amino-acids:step-2 | 2. Glycine exception

Glycine has two hydrogens at its alpha carbon and is achiral. Its neutral net charge near suitable pH does not mean it lacks charged sites.

Displayed molecule ID: glycine

{"highlights": {"prin": [1], "sub": [0, 4]}, "revealName": true}

Screenshot: amino-acids-step-2



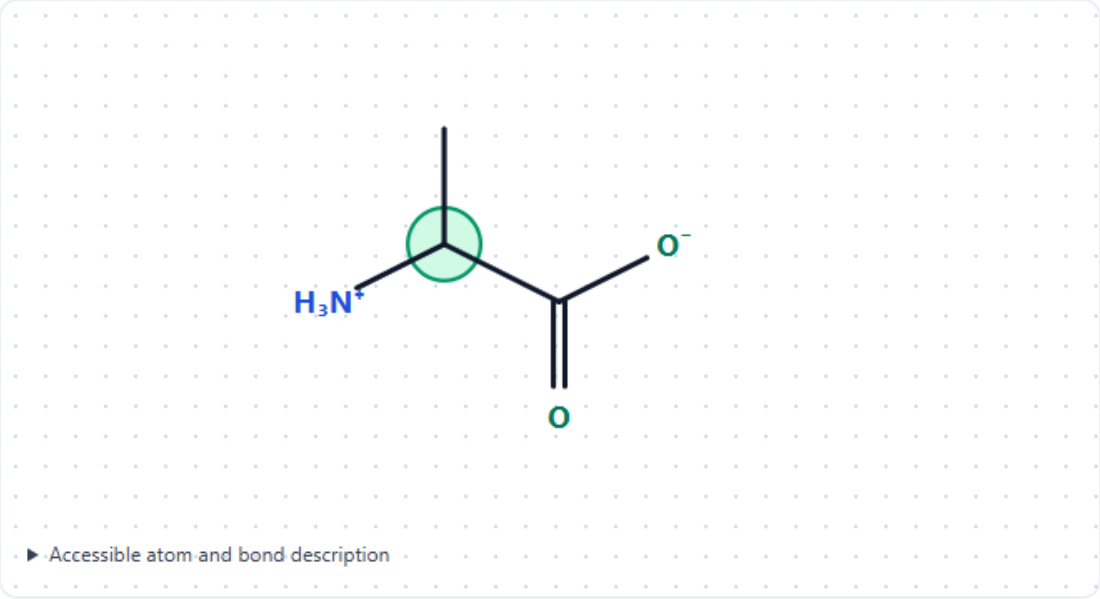
amino-acids:step-3 | 3. Relative configuration

This Fischer projection is the L-alanine arrangement. The carboxyl group is above, side chain below, amino group left. L is not a statement about optical-rotation sign.

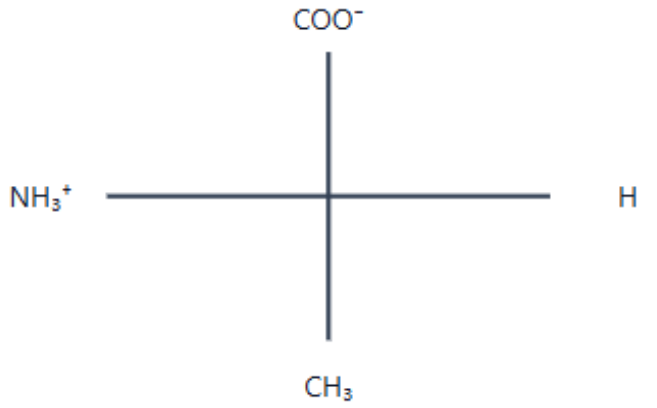
Displayed molecule ID: alanine

{"highlights": {"prin": [1]}, "revealName": false}

Screenshot: amino-acids-step-3



Screenshot: amino-acids-step-3-diagram



L-alanine Fischer projection; the horizontal bonds point toward you.

{"diagramSpecification": {"type": "fischer", "top": "COO⁻", "bottom": "CH₃", "left": "NH₃⁺", "right": "H", "caption": "L-alanine Fischer projection; the horizontal bonds point toward you."}}

peptides | Peptide bonds and protein structure

An amide links amino-acid residues

peptides:rule | Core Rules

A peptide bond joins one residue's carbonyl carbon to the next residue's nitrogen. Resonance makes the peptide group approximately planar with restricted C–N rotation. Sequence is conventionally read N-terminus → C-terminus. Primary sequence supports secondary structures, tertiary folding, and sometimes quaternary assembly.

peptides:cue | Memory Cues

N to C is the reading direction. Peptide C–N is the restricted bond; adjacent backbone single bonds supply much of the flexibility.

peptides:trap | Common Mistakes / MCAT Traps

A peptide bond is not a C–C bond between two alpha carbons. Peptide formation is thermodynamically and kinetically enabled by activation in cells; plain amino acids in water do not rapidly polymerize by themselves.

peptides:app | MCAT Applications

Hydrogen bonding organizes alpha helices and beta sheets; hydrophobic effects and side-chain interactions help stabilize folded proteins.

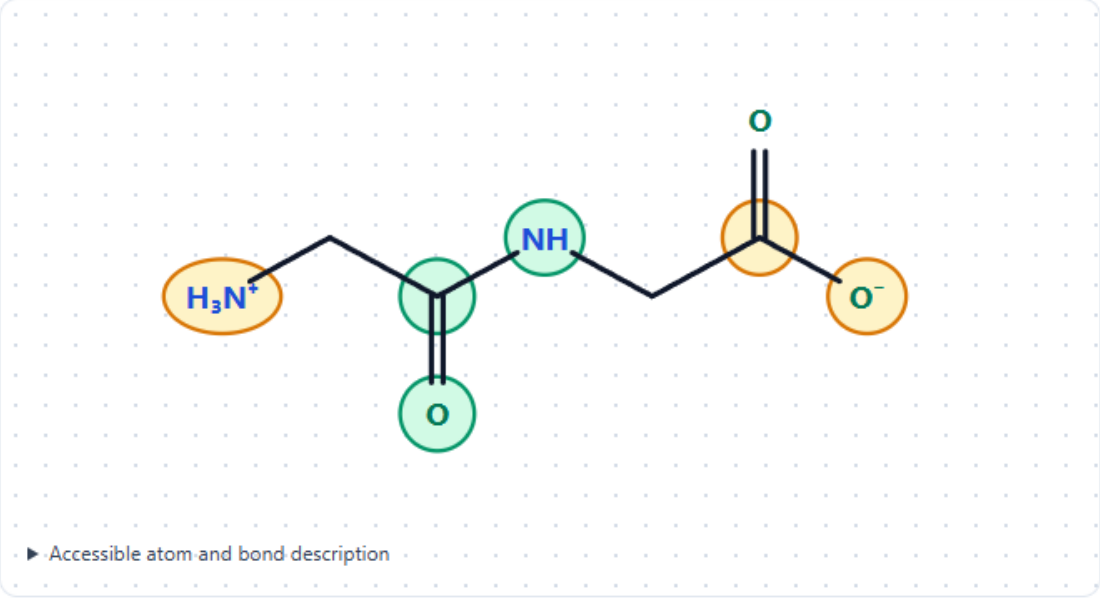
peptides:step-1 | 1. Find the amide linkage

The internal nitrogen is bonded directly to the preceding carbonyl carbon. The terminal amino and carboxyl groups remain available for ionization.

Displayed molecule ID: glycylglycine

{"highlights": {"prin": [2, 3, 4], "sub": [0, 6, 8]}, "revealName": true}

Screenshot: peptides-step-1



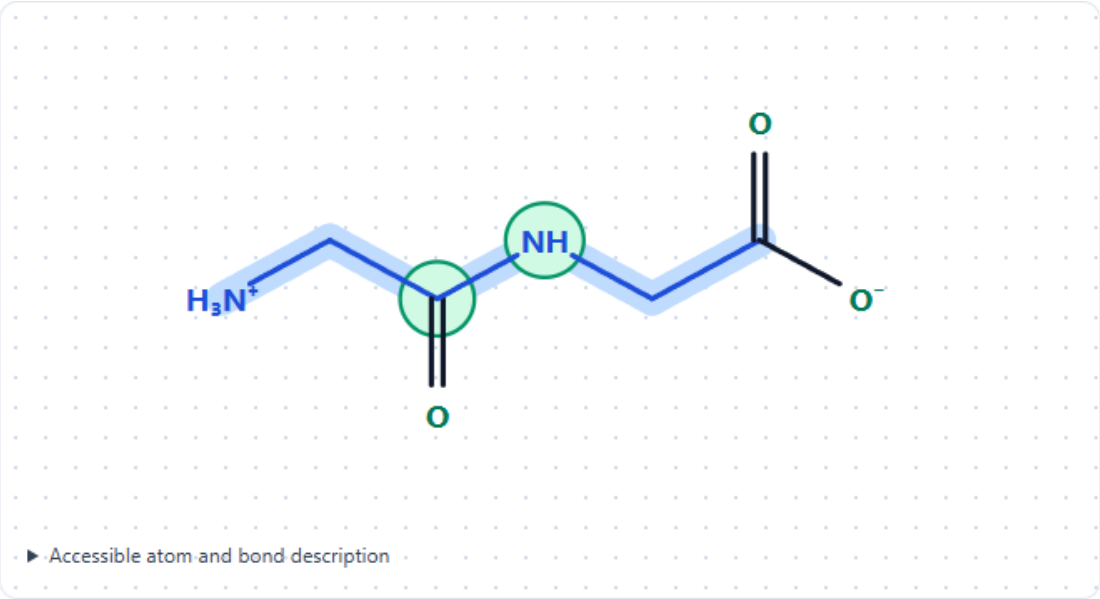
peptides:step-2 | 2. Trace the backbone

Follow N–Cα–C(=O)–N–Cα–C(=O). The peptide C–N link is approximately planar because the N lone pair participates in resonance.

Displayed molecule ID: glycylglycine

{"highlights": {"path": [0, 1, 2, 4, 5, 6], "prin": [2, 4]}, "revealName": false}

Screenshot: peptides-step-2



synthesis | Strecker and Gabriel amino-acid strategies

Track where the carboxyl carbon comes from

synthesis:rule | Core Rules

In the Strecker route, an aldehyde reacts through an imine/iminium stage with ammonia and cyanide to form an alpha-aminonitrile. Nitrile hydrolysis gives the carboxyl group. A Gabriel–malonic ester route installs protected nitrogen and an alkyl side chain, then hydrolyzes and decarboxylates to an alpha amino acid.

synthesis:cue | Memory Cues

Strecker: nitrile carbon becomes carboxyl carbon. Gabriel–malonate: alkylation builds the side chain; decarboxylation removes one carboxyl group.

synthesis:trap | Common Mistakes / MCAT Traps

Achiral starting materials and conditions usually give a racemic chiral amino-acid product; glycine is achiral. Ordinary Gabriel amine synthesis and the malonic-ester amino-acid variant are related but not identical.

synthesis:app | MCAT Applications

Laboratory racemic synthesis contrasts with enzyme-controlled production of stereochemically selected amino acids.

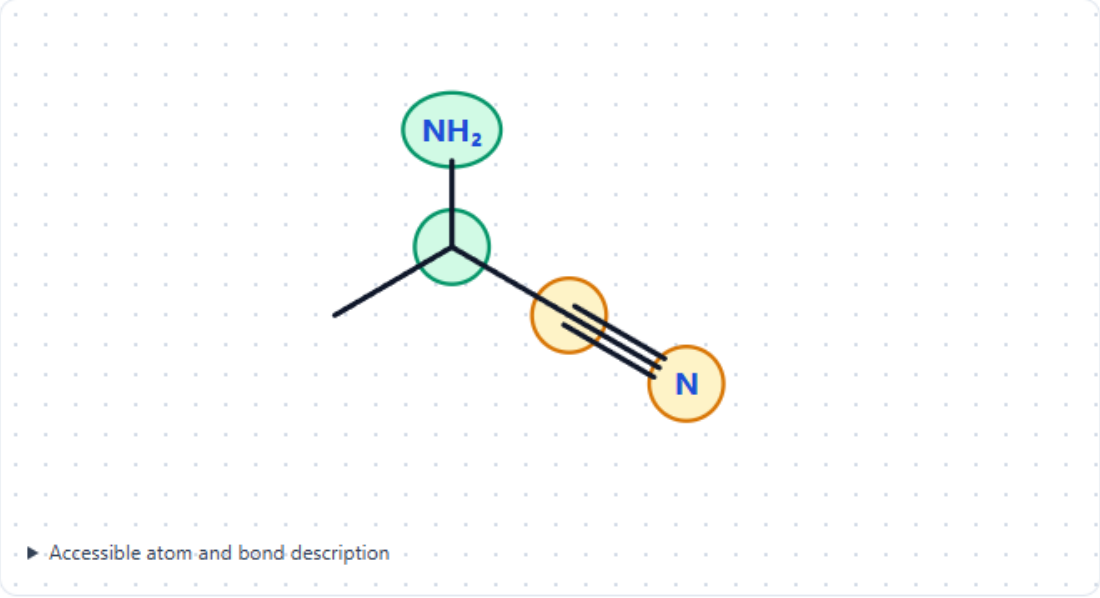
synthesis:step-1 | 1. Aminonitrile intermediate

The central carbon carries an amino group, methyl, H, and a nitrile carbon. This is the carbon skeleton leading to alanine on nitrile hydrolysis.

Displayed molecule ID: aminonitrile

{"highlights": {"prin": [1, 4], "sub": [2, 3]}, "revealName": true}

Screenshot: synthesis-step-1



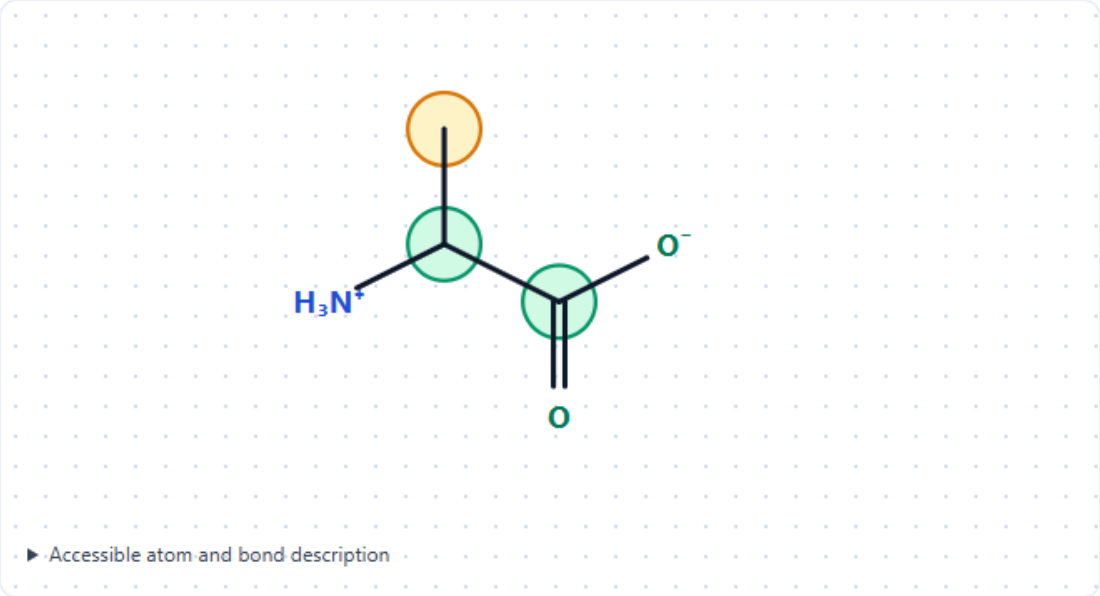
synthesis:step-2 | 2. Carbon bookkeeping

The nitrile carbon becomes the amino acid's carboxyl carbon. The original aldehyde carbon becomes its alpha carbon; the methyl side chain is retained. Protonation state then depends on pH.

Displayed molecule ID: alanine

{"highlights": {"prin": [1, 2], "sub": [5]}, "revealName": true}

Screenshot: synthesis-step-2



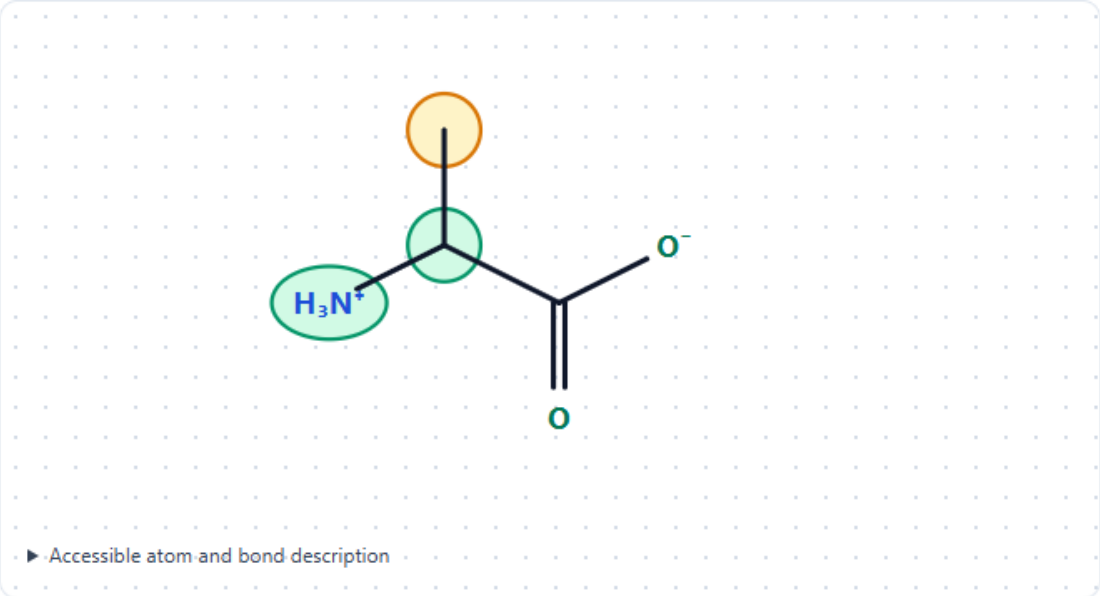
synthesis:step-3 | 3. Alternative protected route

In the Gabriel–malonic ester amino-acid variant, phthalimide supplies protected nitrogen, malonate chemistry enables side-chain alkylation, and hydrolysis/deprotection plus decarboxylation yields the amino acid. This is a route overview, not a laboratory protocol.

Displayed molecule ID: alanine

{"highlights": {"prin": [0, 1], "sub": [5]}, "revealName": false}

Screenshot: synthesis-step-3



phosphorus | Phosphates and phosphoryl transfer

Phosphoesters differ from phosphoanhydrides

phosphorus:rule | Core Rules

A phosphate ester contains P–O–C; a phosphodiester has two such ester links. A phosphoanhydride contains P–O–P. Phosphoric acid has successive dissociations, so charge depends on pH. Formal P=O Lewis drawings are bookkeeping models of delocalized bonding.

phosphorus:cue | Memory Cues

DNA/RNA backbone: phosphodiester. Between ATP phosphate groups: phosphoanhydride. Approximate aqueous phosphoric-acid pKa values are 2.1, 7.2, and 12.3, so the $H_2PO_4^-/HPO_4^{2-}$ pair can buffer near neutral pH; temperature and ionic conditions shift these values.

phosphorus:trap | Common Mistakes / MCAT Traps

Breaking a bond requires energy. Favorable ATP hydrolysis reflects the overall free-energy difference between reactants and solvated products, including resonance, electrostatics, and solvation; energy is not released simply by breaking one bond.

phosphorus:app | MCAT Applications

Phosphoryl transfer can couple reactions. Nucleic-acid synthesis releases pyrophosphate, whose subsequent hydrolysis can help favor polymerization.

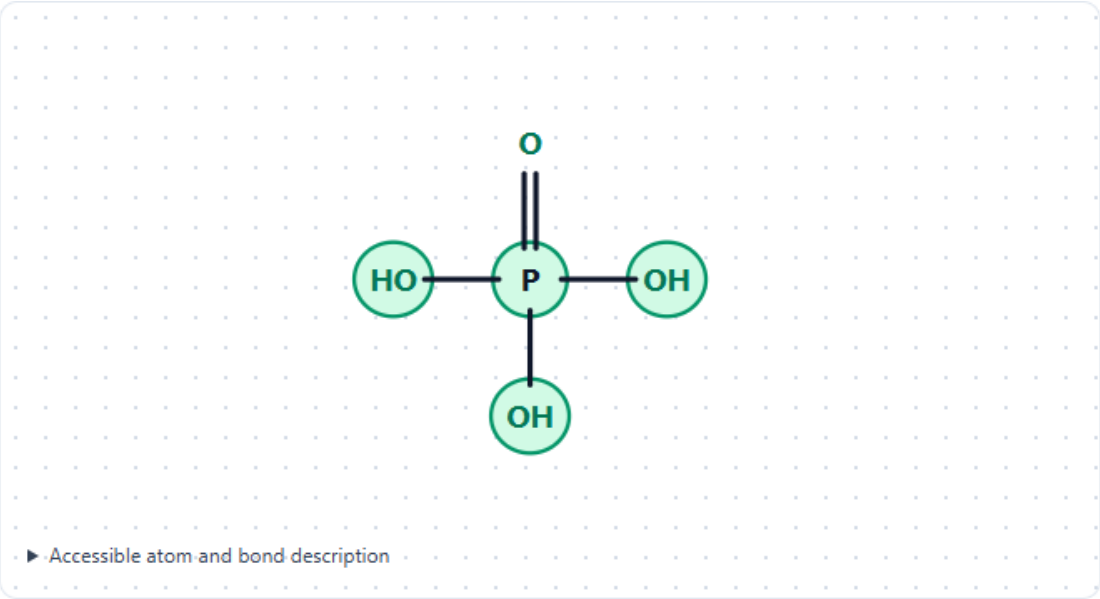
phosphorus:step-1 | 1. Acidic phosphate

The neutral phosphoric-acid representation has three OH groups. Sequential deprotonation gives different phosphate ions; this drawing is not the predominant form at every pH.

Displayed molecule ID: phosphoricacid

{"highlights": {"prin": [0, 2, 3, 4]}, "revealName": true}

Screenshot: phosphorus-step-1



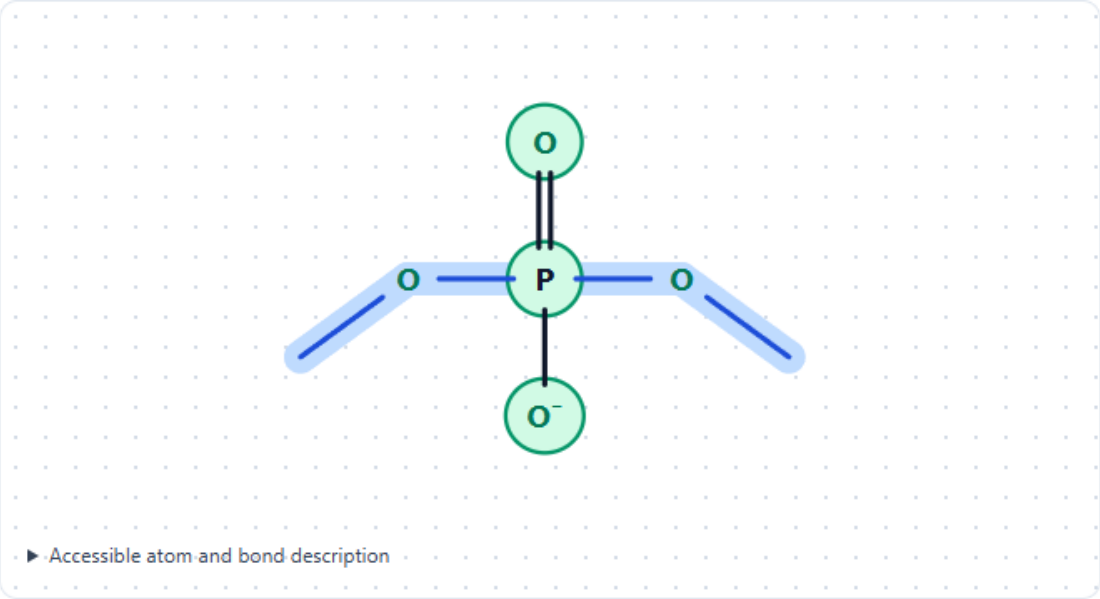
phosphorus:step-2 | 2. Phosphodiester model

Dimethyl phosphate illustrates two P–O–C connections and a negative charge. The nucleic-acid backbone replaces the methyl groups with sugar-derived groups.

Displayed molecule ID: dimethylphosphate

{"highlights": {"path": [5, 2, 0, 3, 6], "prin": [0, 1, 4]}, "revealName": true}

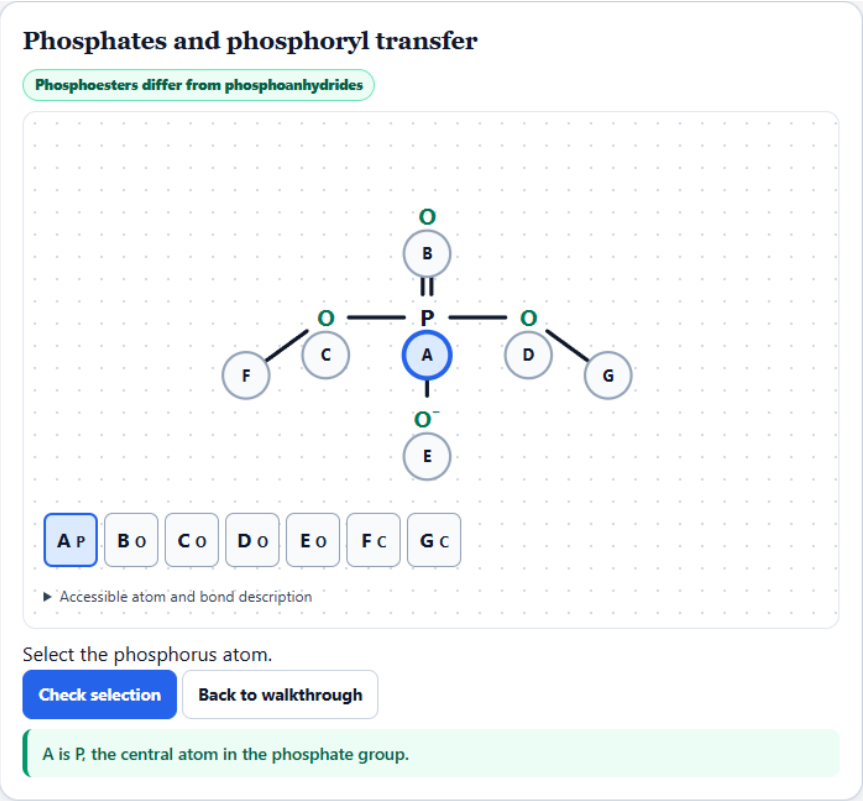
Screenshot: phosphorus-step-2



Embedded challenge: Select the phosphorus atom.

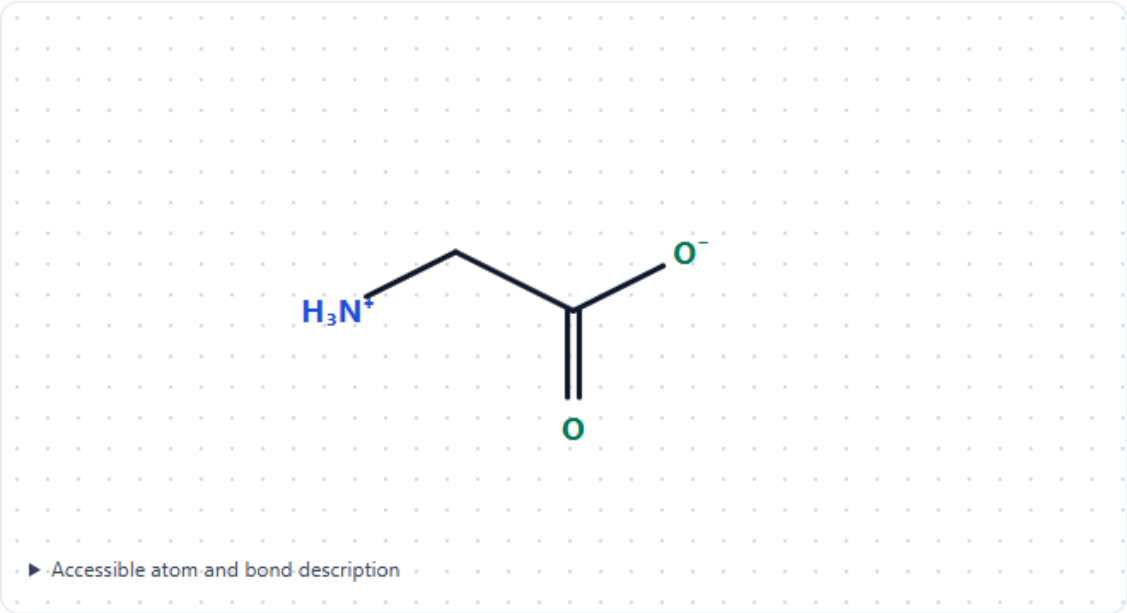
{"prompt": "Select the phosphorus atom.", "answer": [0], "explanation": "A is P, the central atom in the phosphate group."}

Screenshot: phosphorus-step-2-challenge



glycine-charge | Charge and isoelectric point

Screenshot: glycine-charge-stage-1-molecule



glycine-charge:stage-1 | What is the net formal charge of the displayed zwitterion?

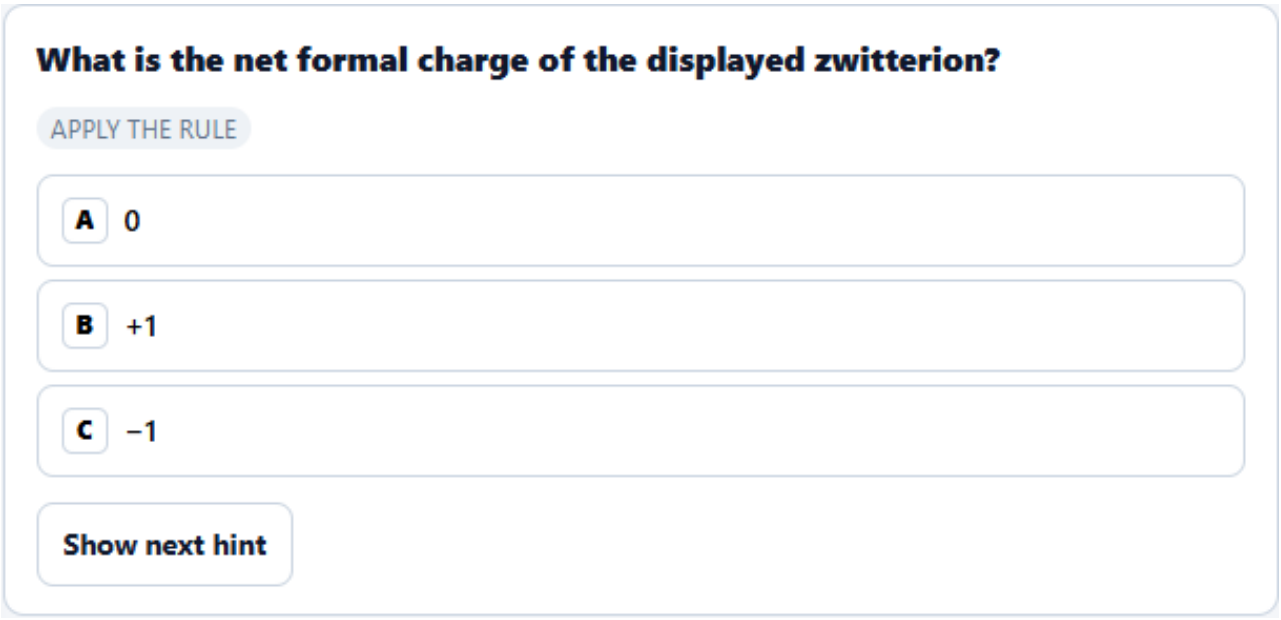
Answer type: choice

Option 0 (A): 0

Option 1 (B): +1

Option 2 (C): −1

Screenshot: glycine-charge-stage-1-question



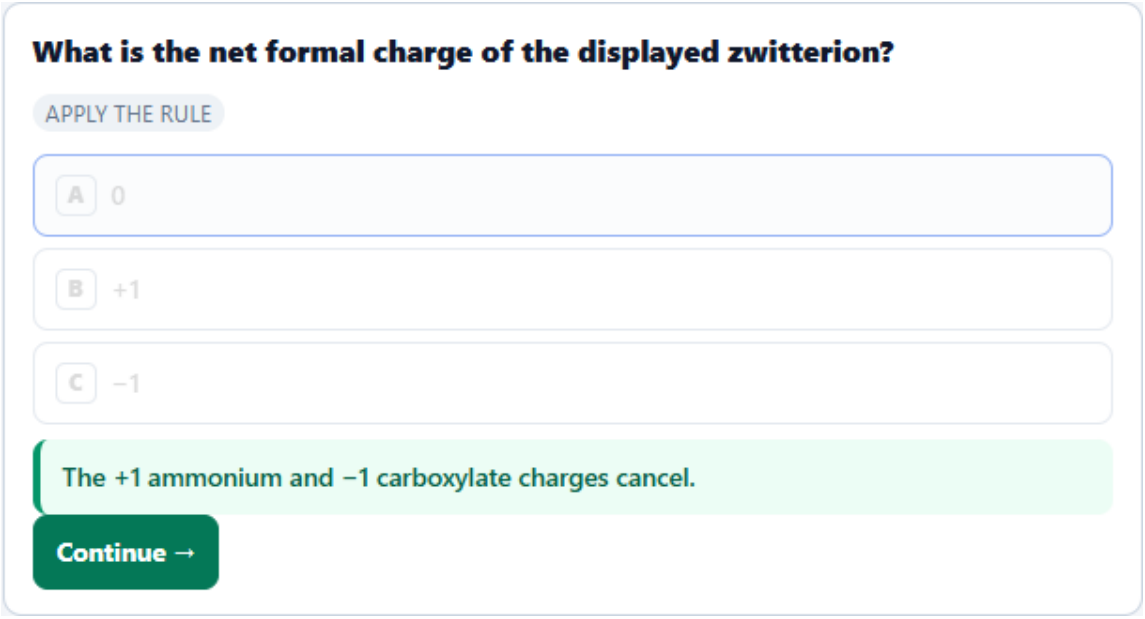
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Add the labeled formal charges.

Hint 2: Net zero does not mean no charged sites.

Explanation: The +1 ammonium and −1 carboxylate charges cancel.

Screenshot: glycine-charge-stage-1-feedback



glycine-charge:stage-2 | Why is glycine achiral?

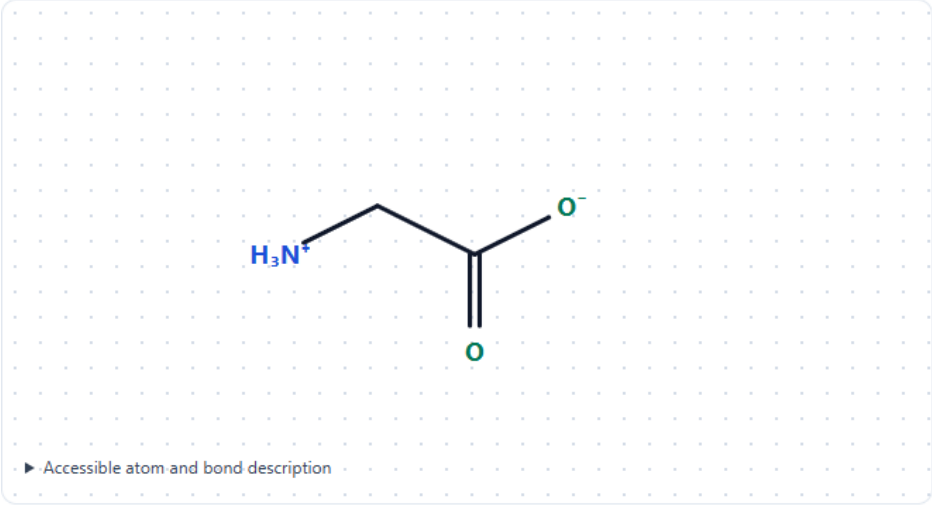
Answer type: choice

Option 0 (A): Its alpha carbon has two H substituents

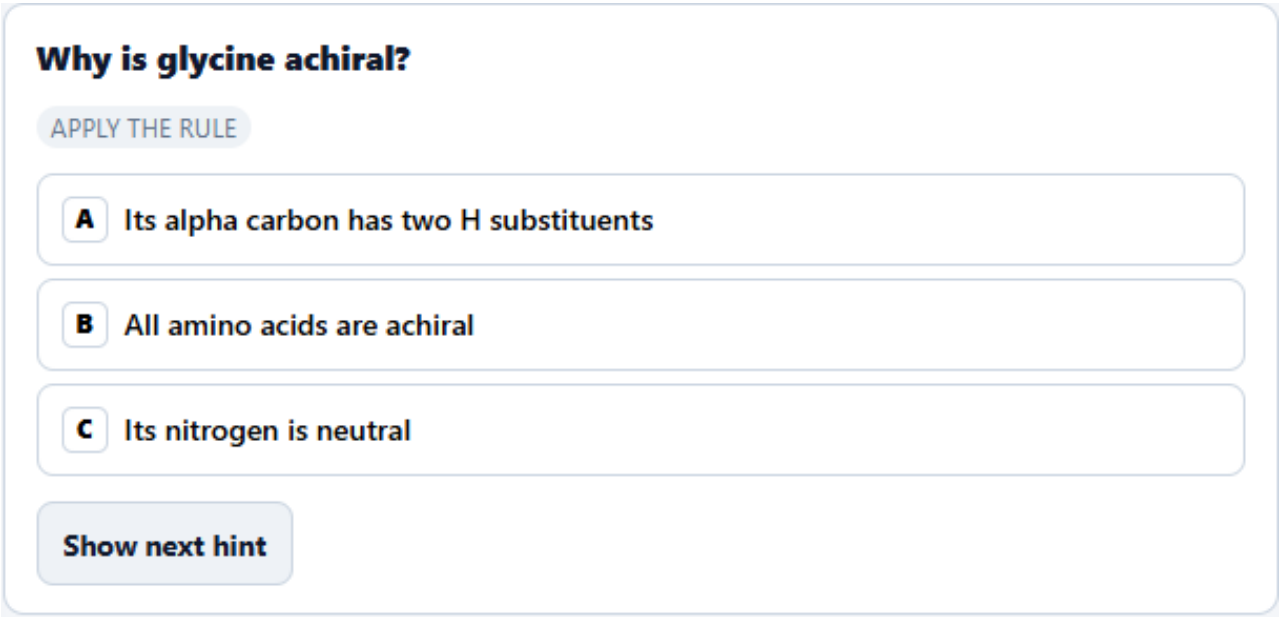
Option 1 (B): All amino acids are achiral

Option 2 (C): Its nitrogen is neutral

Screenshot: glycine-charge-stage-2-molecule



Screenshot: glycine-charge-stage-2-question



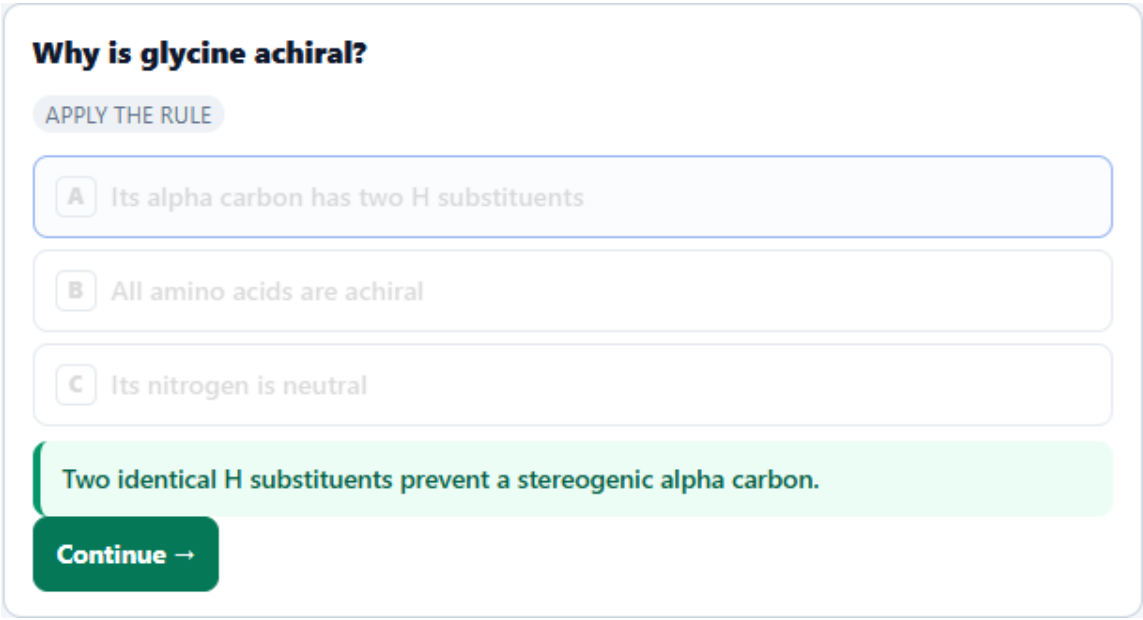
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Inspect the implicit hydrogens.

Hint 2: The side chain itself is H.

Explanation: Two identical H substituents prevent a stereogenic alpha carbon.

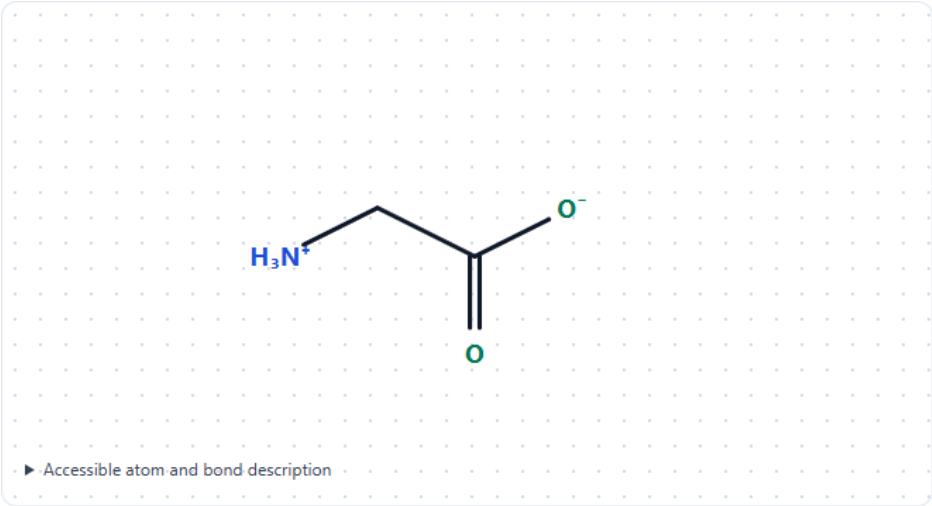
Screenshot: glycine-charge-stage-2-feedback



glycine-charge:stage-3 | Using pKa values 2.3 and 9.6 for this neutral-side-chain model, calculate pI.

Answer type: number | unit: pH units

Screenshot: glycine-charge-stage-3-molecule



Screenshot: glycine-charge-stage-3-question

Using pKa values 2.3 and 9.6 for this neutral-side-chain model, calculate pI.

CALCULATE AND INTERPRET

Numeric answer (pH units)

Check answer

Show next hint

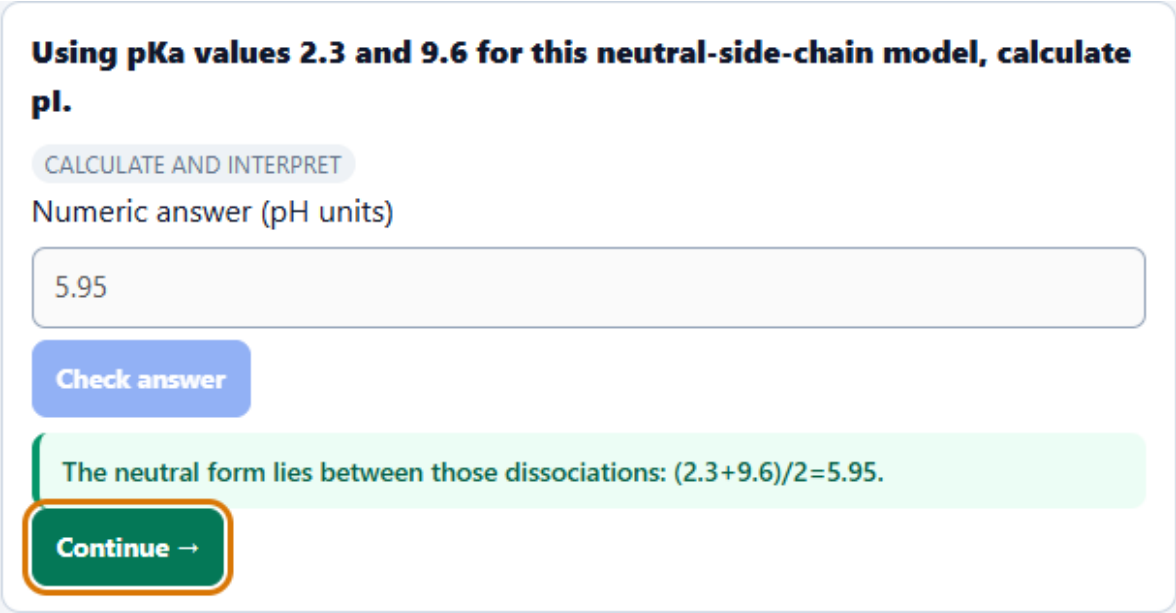
{"authorAnswer": 5.95, "tolerance": 0.005, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the pKa pair bracketing net zero.

Hint 2: Average the two supplied numbers.

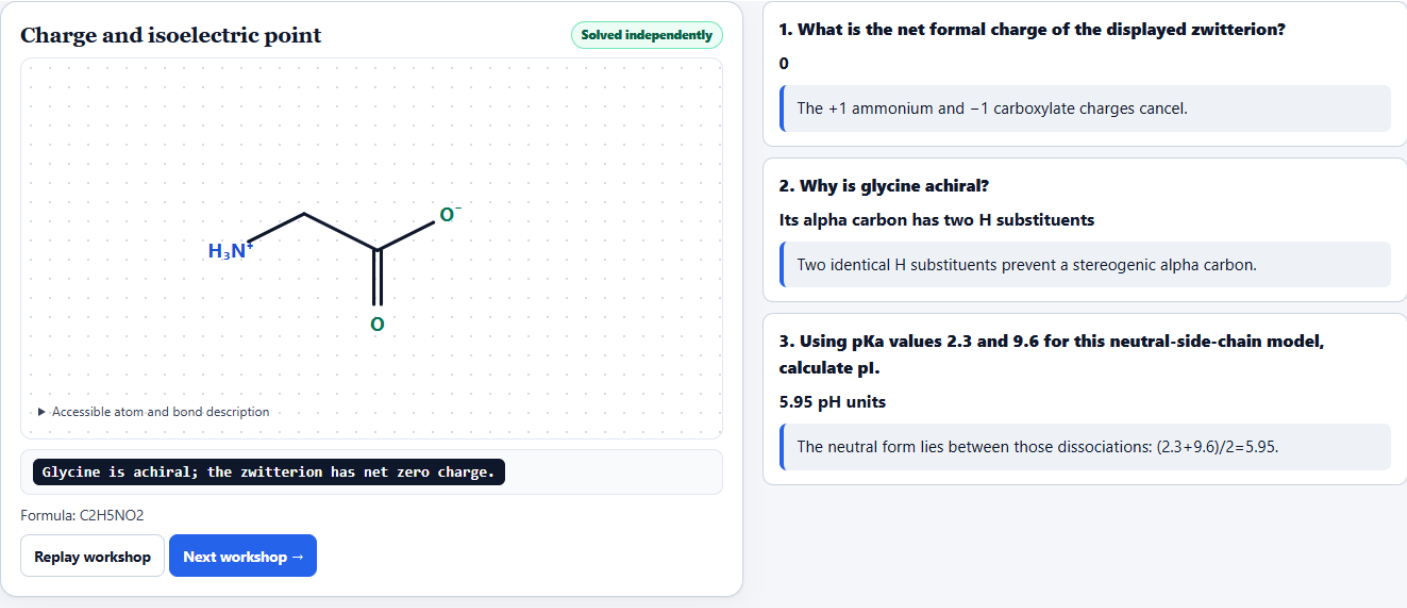
Explanation: The neutral form lies between those dissociations: (2.3+9.6)/2=5.95.

Screenshot: glycine-charge-stage-3-feedback



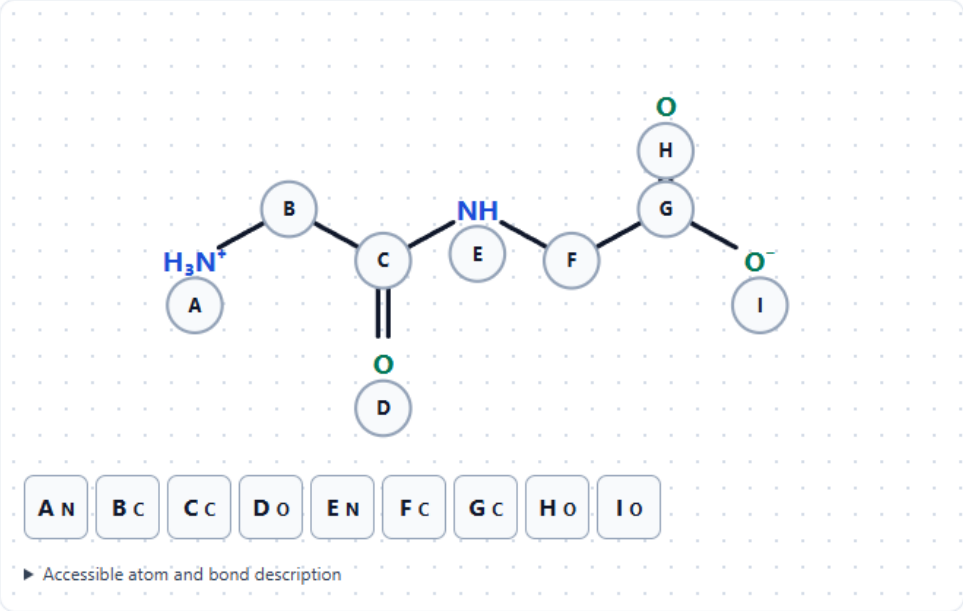
Completion summary: Glycine is achiral; the zwitterion has net zero charge.

Screenshot: glycine-charge-complete



peptide-link | Read a peptide backbone

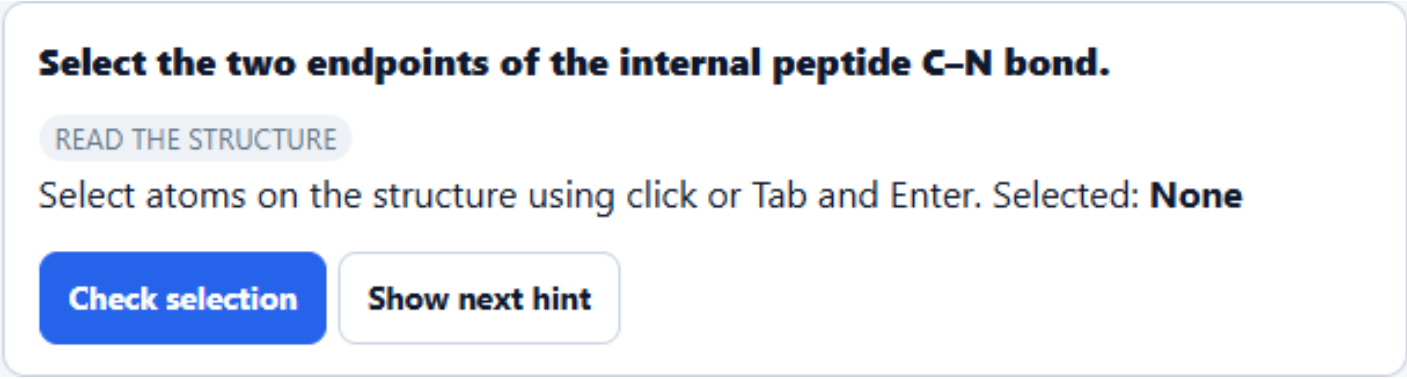
Screenshot: peptide-link-stage-1-molecule



peptide-link:stage-1 | Select the two endpoints of the internal peptide C–N bond.

Answer type: atoms

Screenshot: peptide-link-stage-1-question



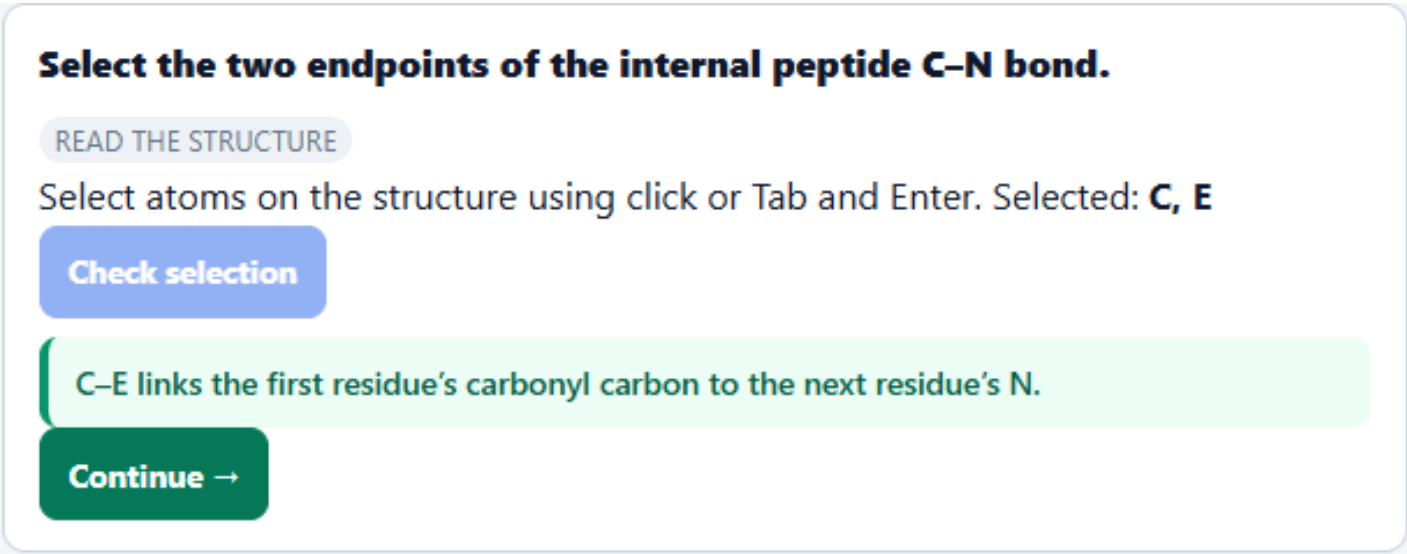
{"authorAnswer": [2, 4], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 4]}}

Hint 1: Find C(=O)–N inside the molecule.

Hint 2: Exclude the terminal ammonium nitrogen.

Explanation: C–E links the first residue's carbonyl carbon to the next residue's N.

Screenshot: peptide-link-stage-1-feedback



peptide-link:stage-2 | Why is rotation around this bond restricted?

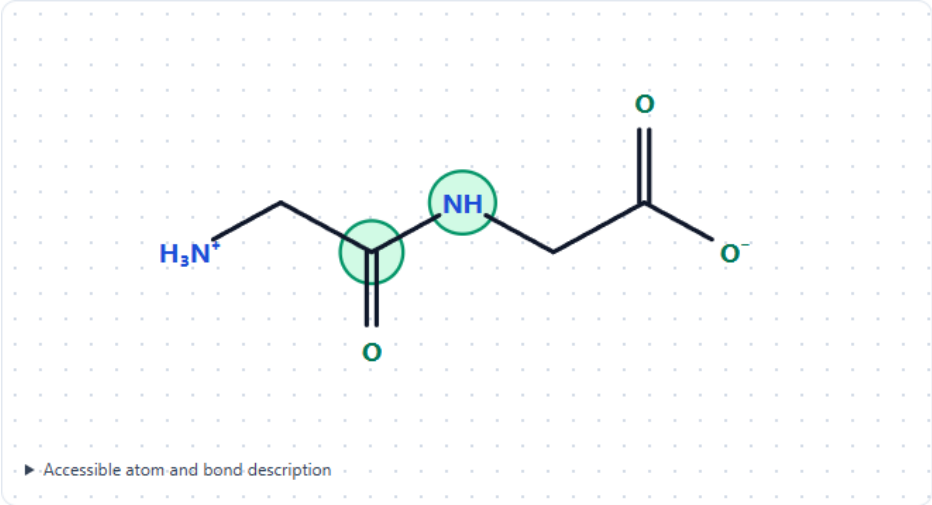
Answer type: choice

Option 0 (A): Amide resonance gives partial double-bond character

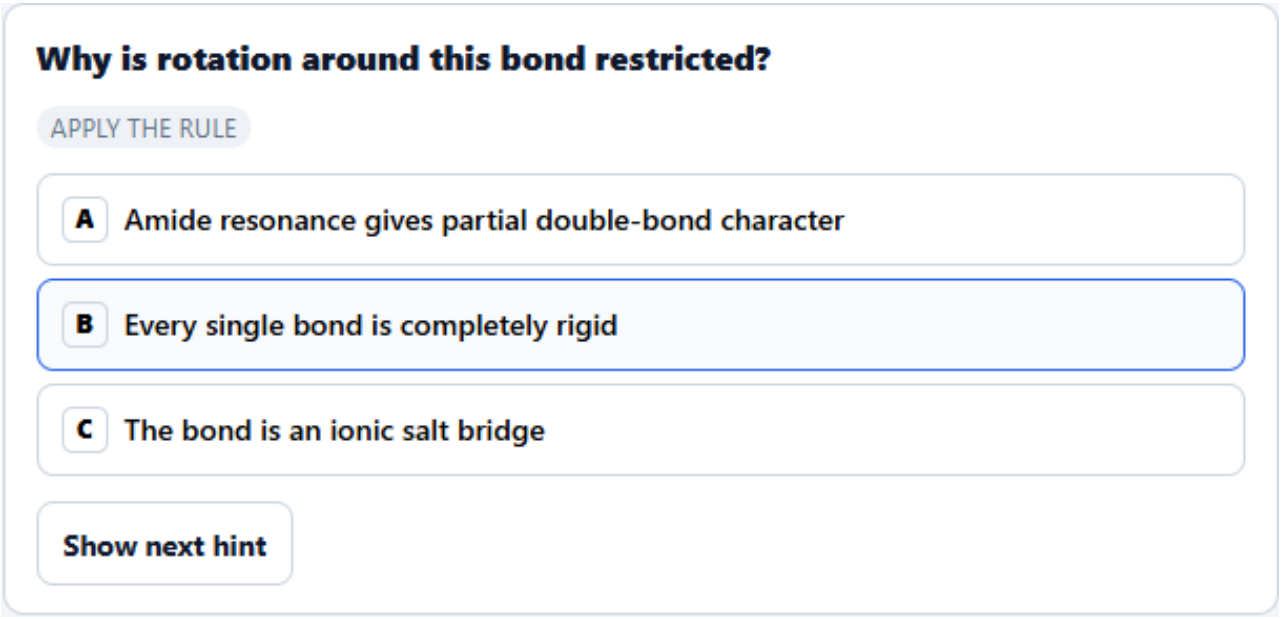
Option 1 (B): Every single bond is completely rigid

Option 2 (C): The bond is an ionic salt bridge

Screenshot: peptide-link-stage-2-molecule



Screenshot: peptide-link-stage-2-question



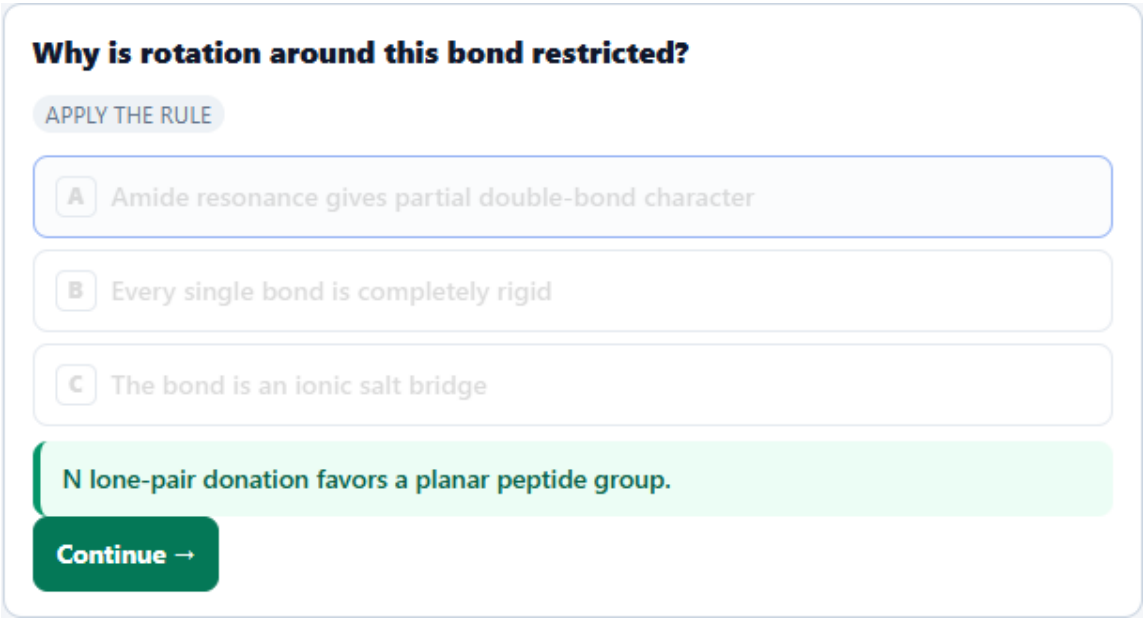
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Apply the amide resonance model.

Hint 2: Aligned p orbitals stabilize the group.

Explanation: N lone-pair donation favors a planar peptide group.

Screenshot: peptide-link-stage-2-feedback



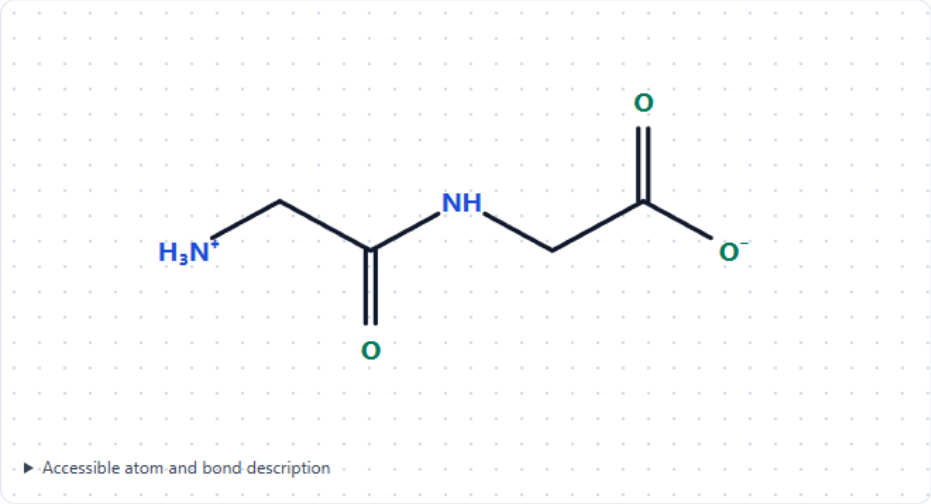
peptide-link:stage-3 | Peptide sequences are read from...

Answer type: choice

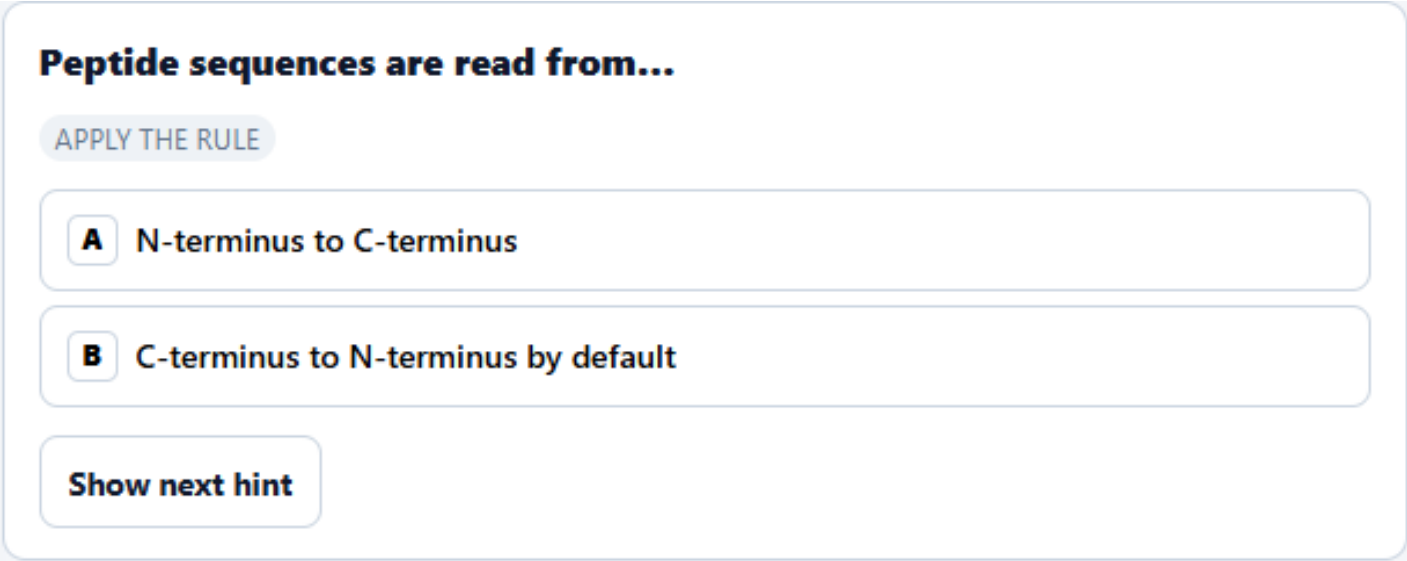
Option 0 (A): N-terminus to C-terminus

Option 1 (B): C-terminus to N-terminus by default

Screenshot: peptide-link-stage-3-molecule



Screenshot: peptide-link-stage-3-question



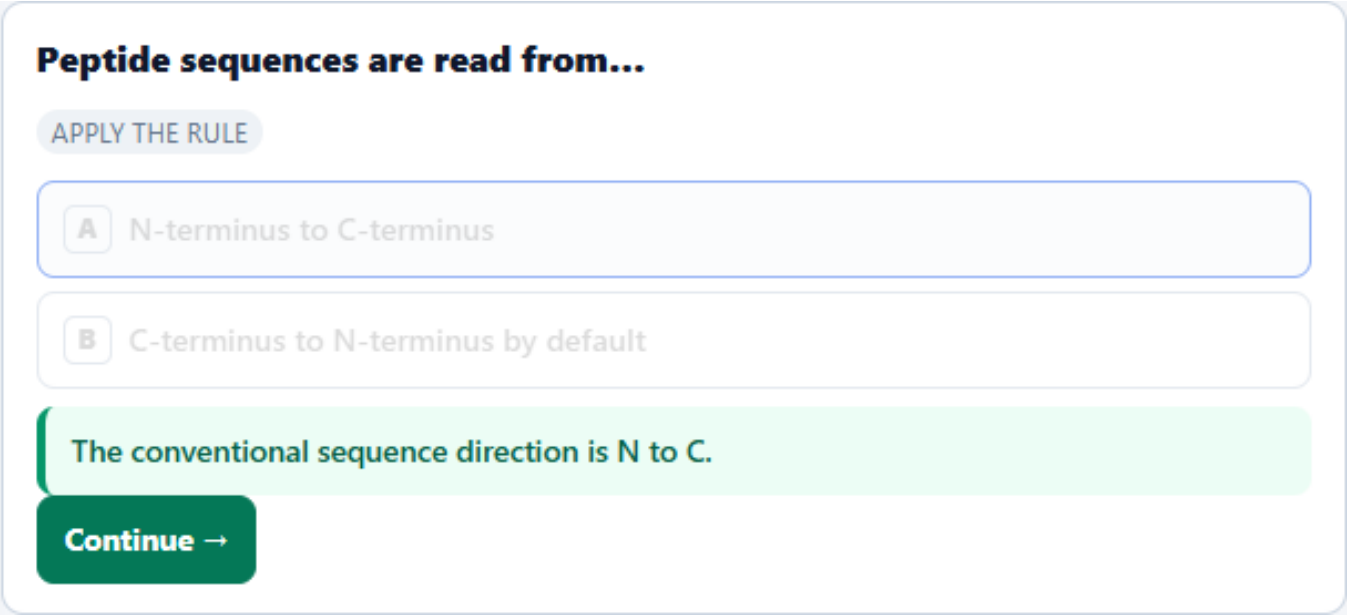
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Find the free terminal amino group.

Hint 2: Start there and follow the backbone.

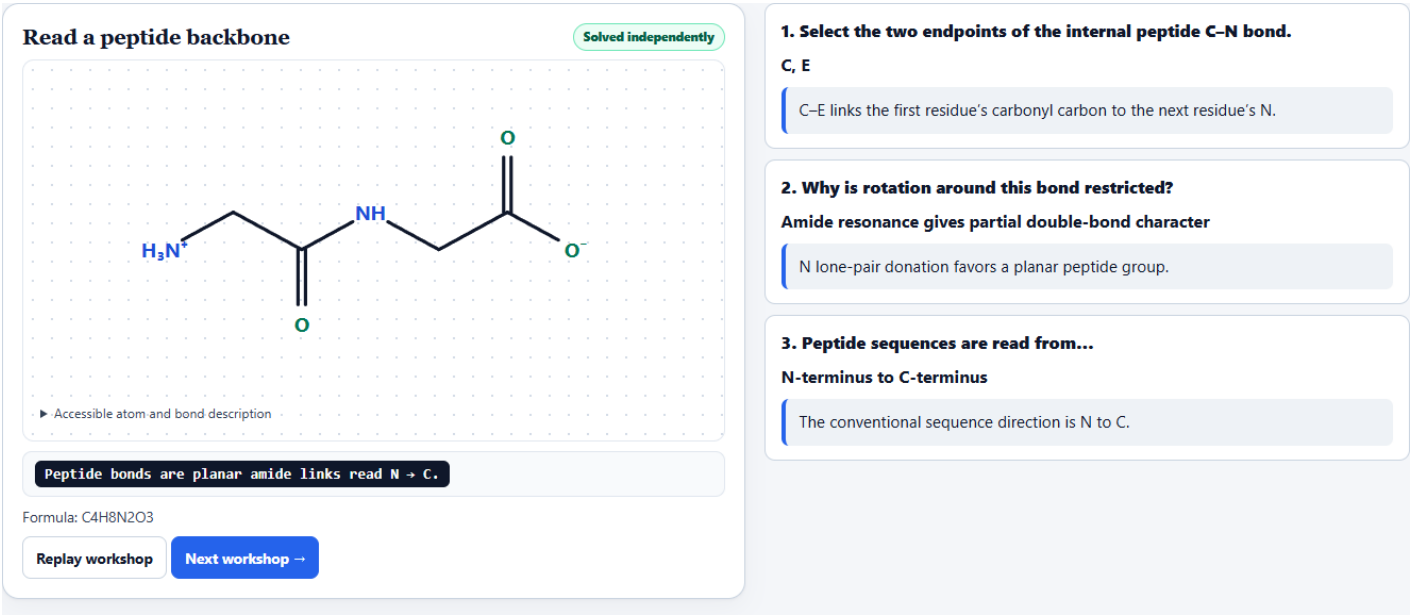
Explanation: The conventional sequence direction is N to C.

Screenshot: peptide-link-stage-3-feedback



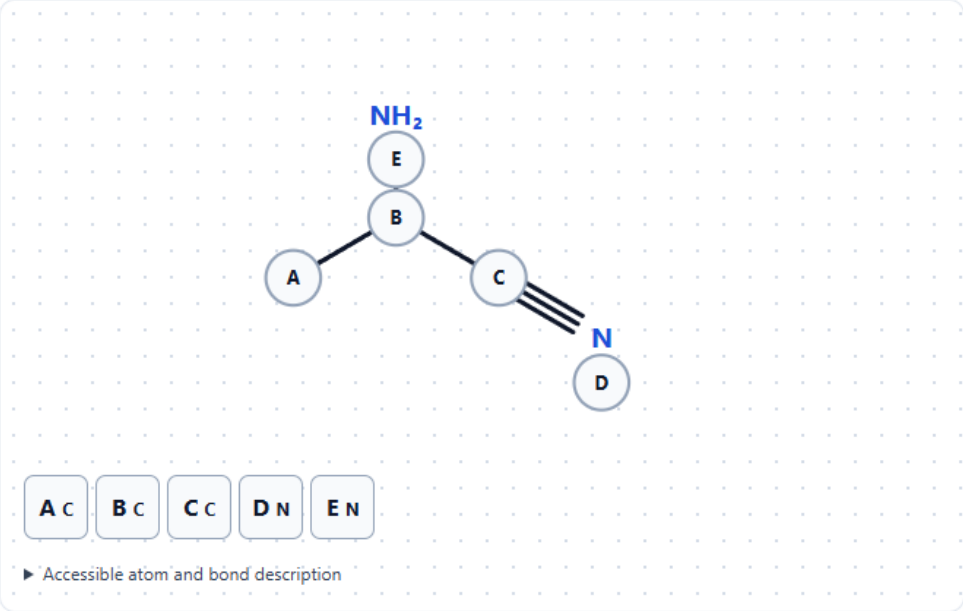
Completion summary: Peptide bonds are planar amide links read N → C.

Screenshot: peptide-link-complete



strecker-map | Track a Strecker intermediate

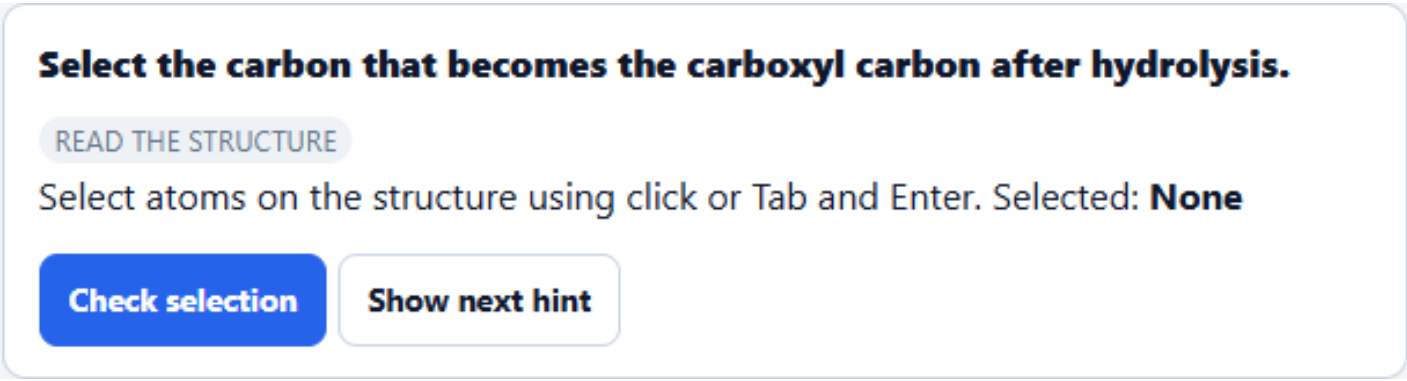
Screenshot: strecker-map-stage-1-molecule



strecker-map:stage-1 | Select the carbon that becomes the carboxyl carbon after hydrolysis.

Answer type: atoms

Screenshot: strecker-map-stage-1-question



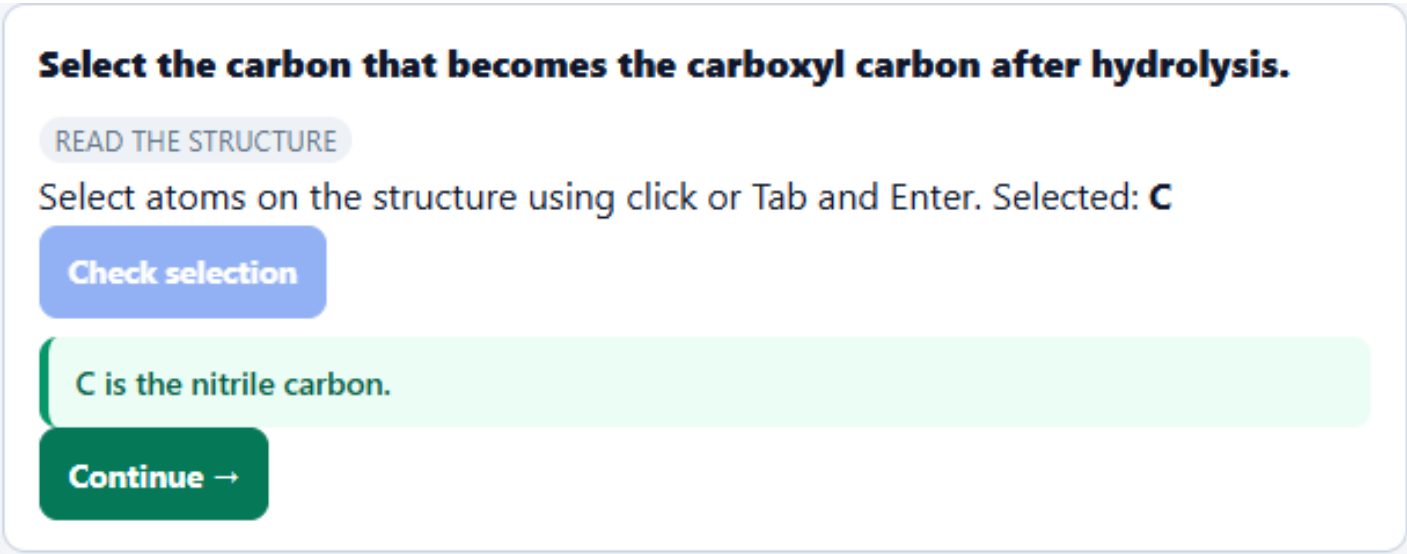
{"authorAnswer": [2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 3]}}

Hint 1: Hydrolysis transforms C≡N into a carboxyl group.

Hint 2: Track the carbon, not the nitrile nitrogen.

Explanation: C is the nitrile carbon.

Screenshot: strecker-map-stage-1-feedback



strecker-map:stage-2 | Which group is retained as the alanine side chain?

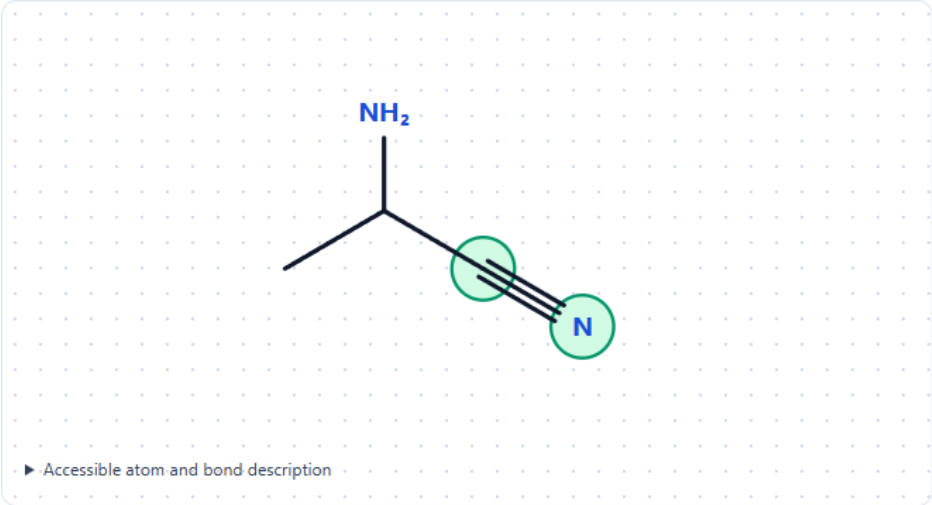
Answer type: choice

Option 0 (A): Methyl

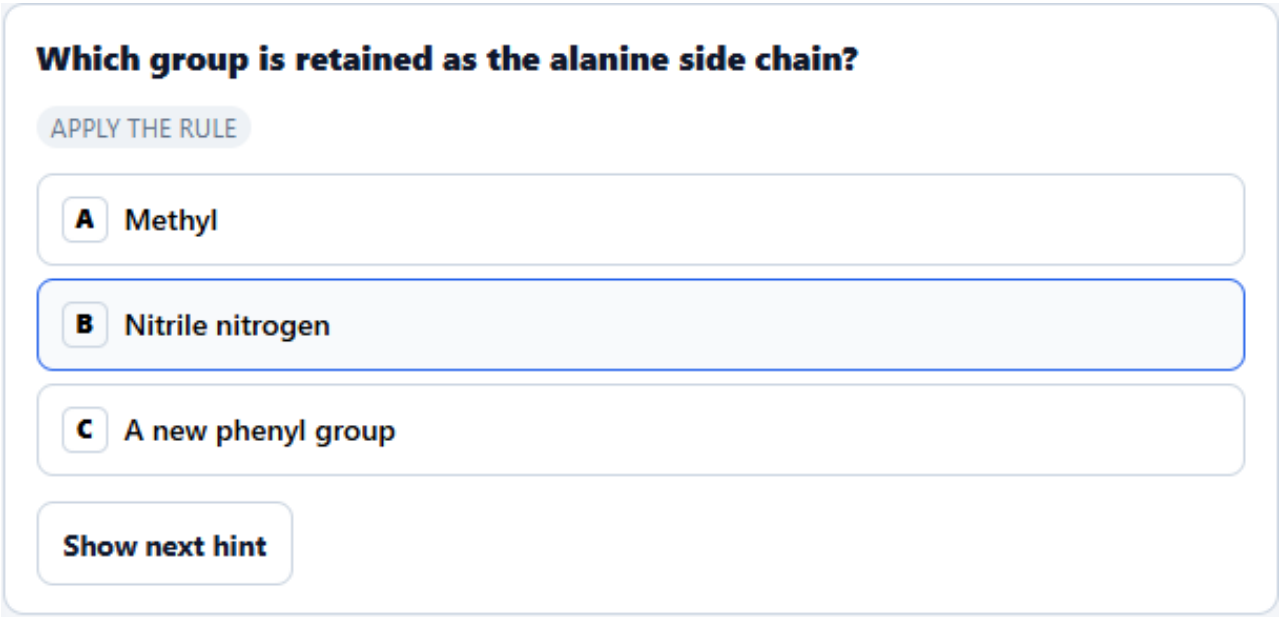
Option 1 (B): Nitrile nitrogen

Option 2 (C): A new phenyl group

Screenshot: strecker-map-stage-2-molecule



Screenshot: strecker-map-stage-2-question



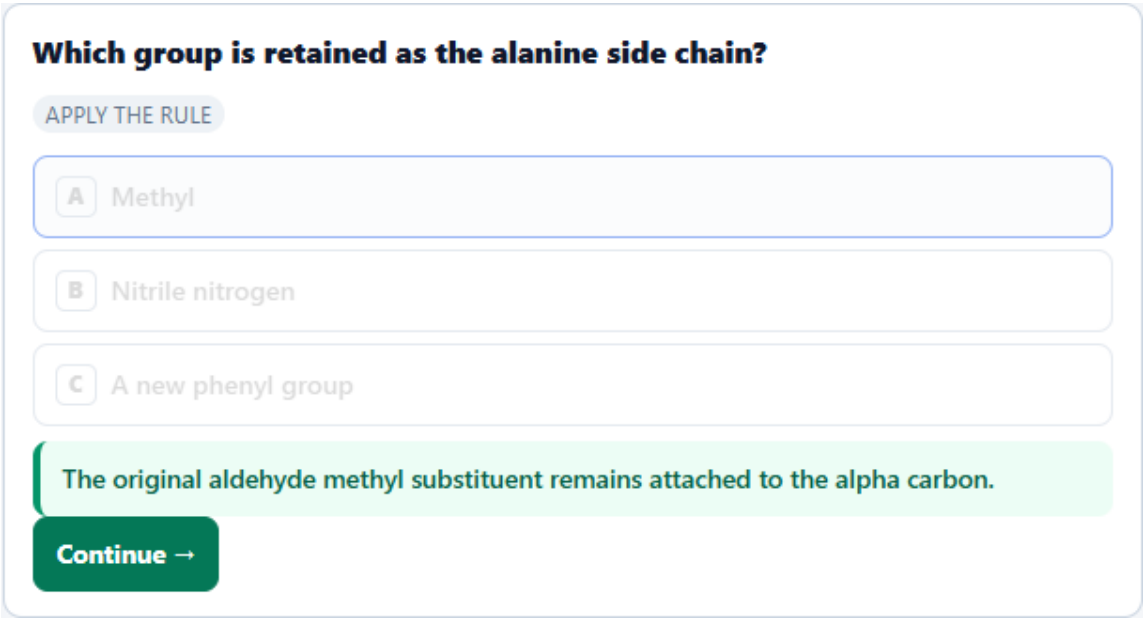
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Follow the carbon skeleton.

Hint 2: No new aromatic group is introduced.

Explanation: The original aldehyde methyl substituent remains attached to the alpha carbon.

Screenshot: strecker-map-stage-2-feedback



strecker-map:stage-3 | Under achiral conditions, the chiral product is generally...

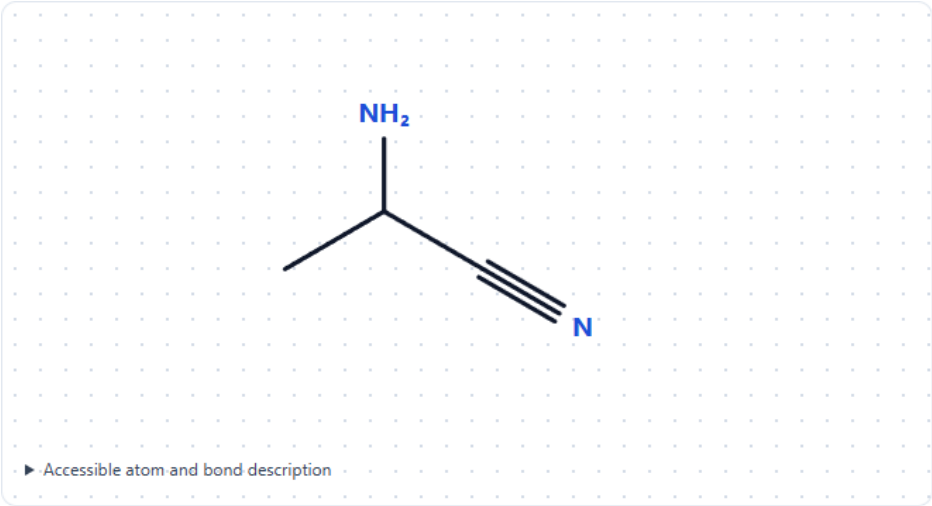
Answer type: choice

Option 0 (A): Racemic

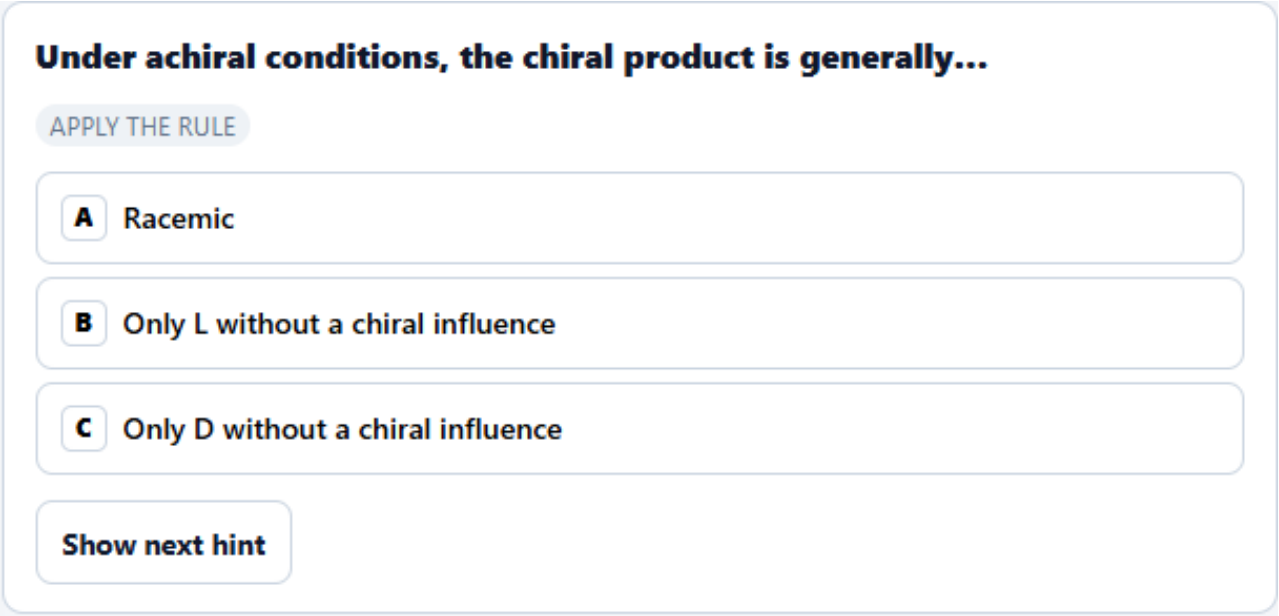
Option 1 (B): Only L without a chiral influence

Option 2 (C): Only D without a chiral influence

Screenshot: strecker-map-stage-3-molecule



Screenshot: strecker-map-stage-3-question



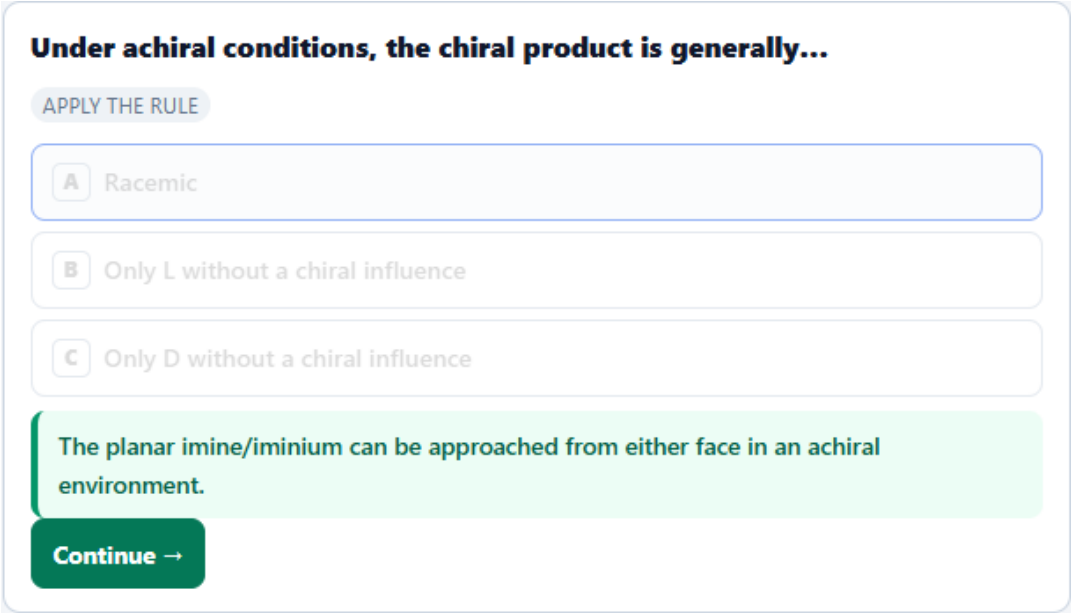
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Ask whether the reaction distinguishes enantiotopic faces.

Hint 2: No chiral catalyst or reagent is specified.

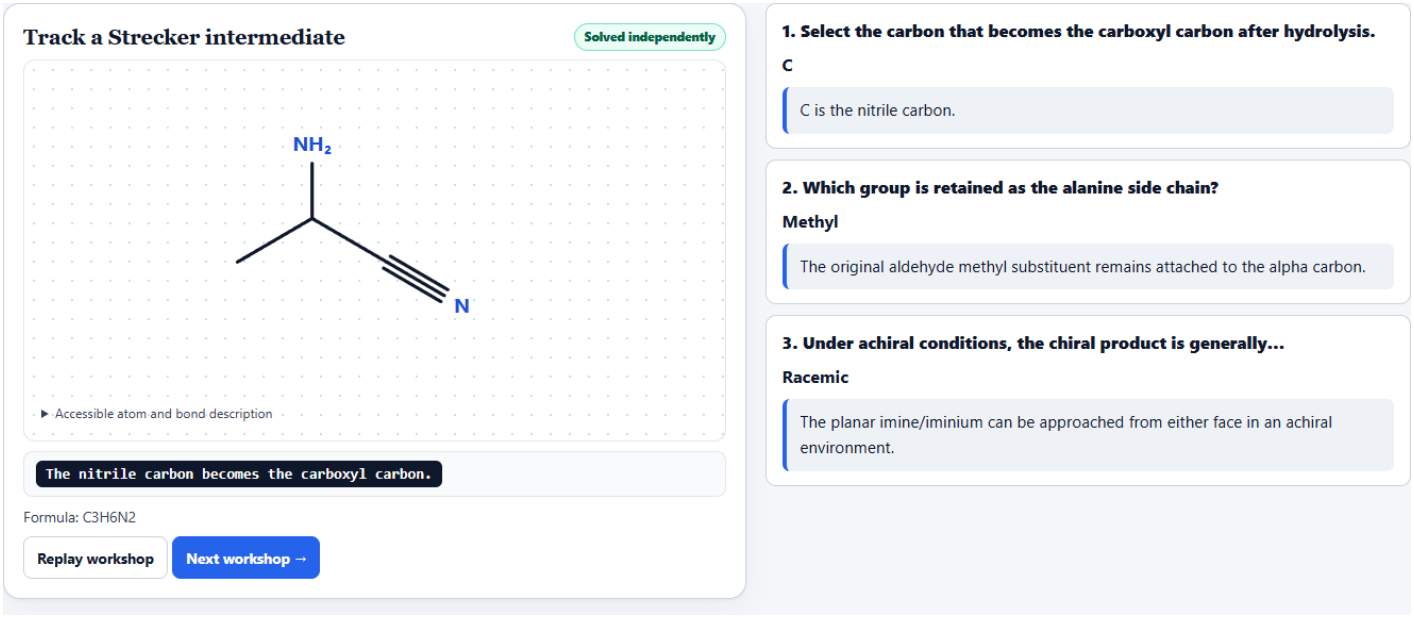
Explanation: The planar imine/iminium can be approached from either face in an achiral environment.

Screenshot: strecker-map-stage-3-feedback



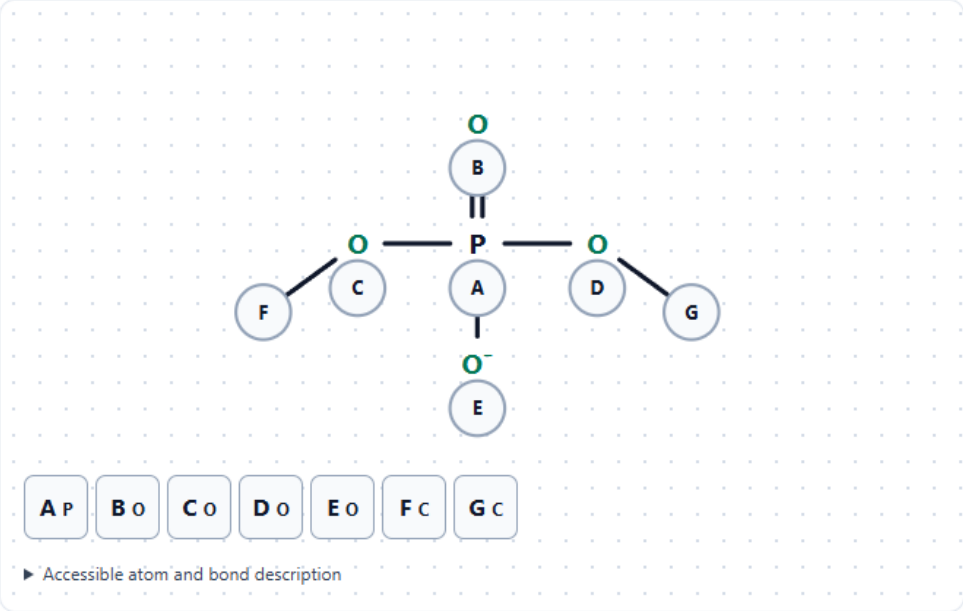
Completion summary: The nitrile carbon becomes the carboxyl carbon.

Screenshot: strecker-map-complete



phosphate-links | Identify phosphorus linkages

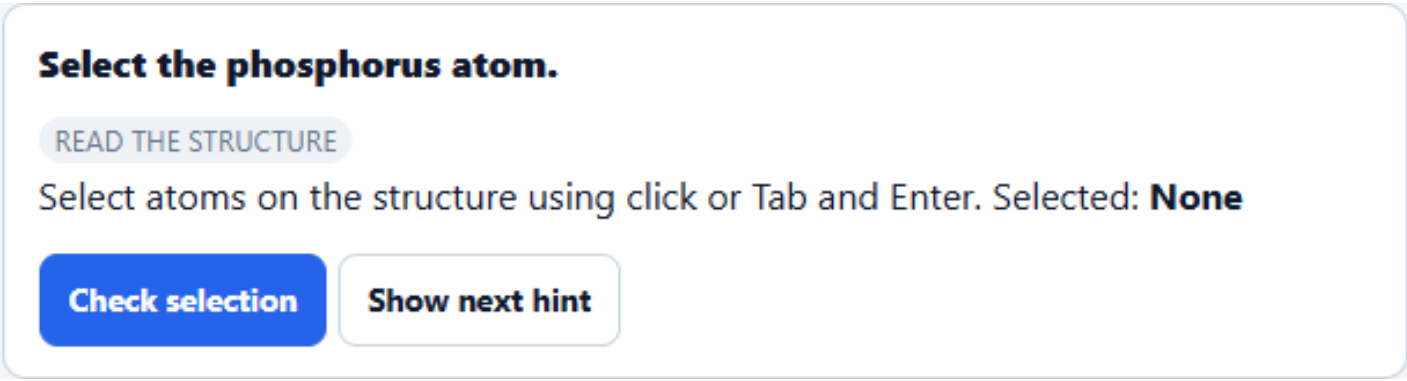
Screenshot: phosphate-links-stage-1-molecule



phosphate-links:stage-1 | Select the phosphorus atom.

Answer type: atoms

Screenshot: phosphate-links-stage-1-question



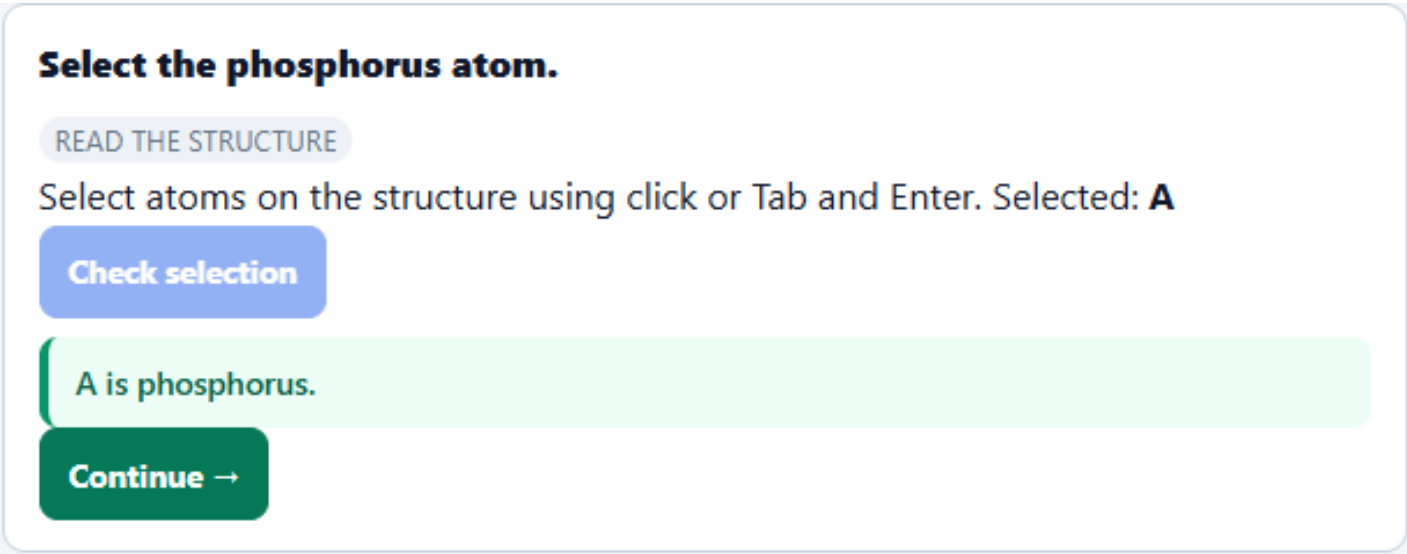
{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0]}}

Hint 1: Read the P label.

Hint 2: It is bonded to four oxygens.

Explanation: A is phosphorus.

Screenshot: phosphate-links-stage-1-feedback



phosphate-links:stage-2 | This structure models a...

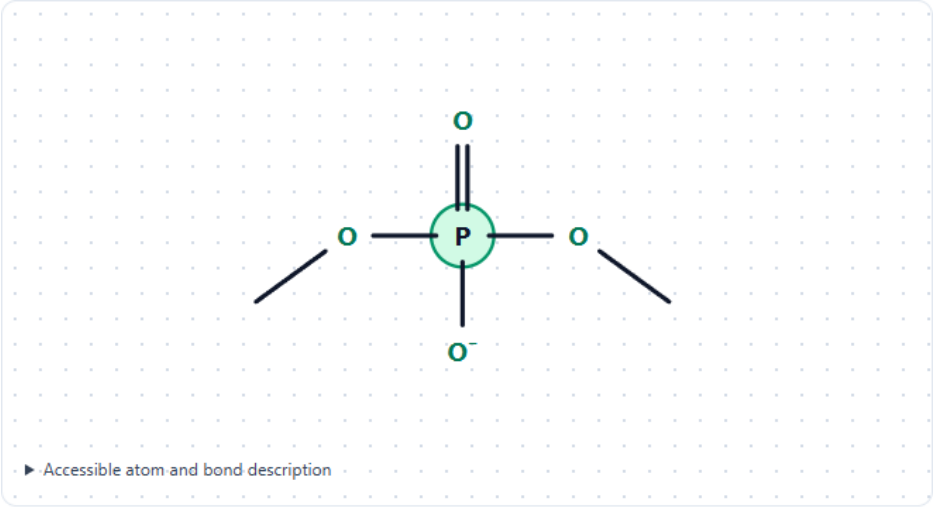
Answer type: choice

Option 0 (A): Phosphodiester

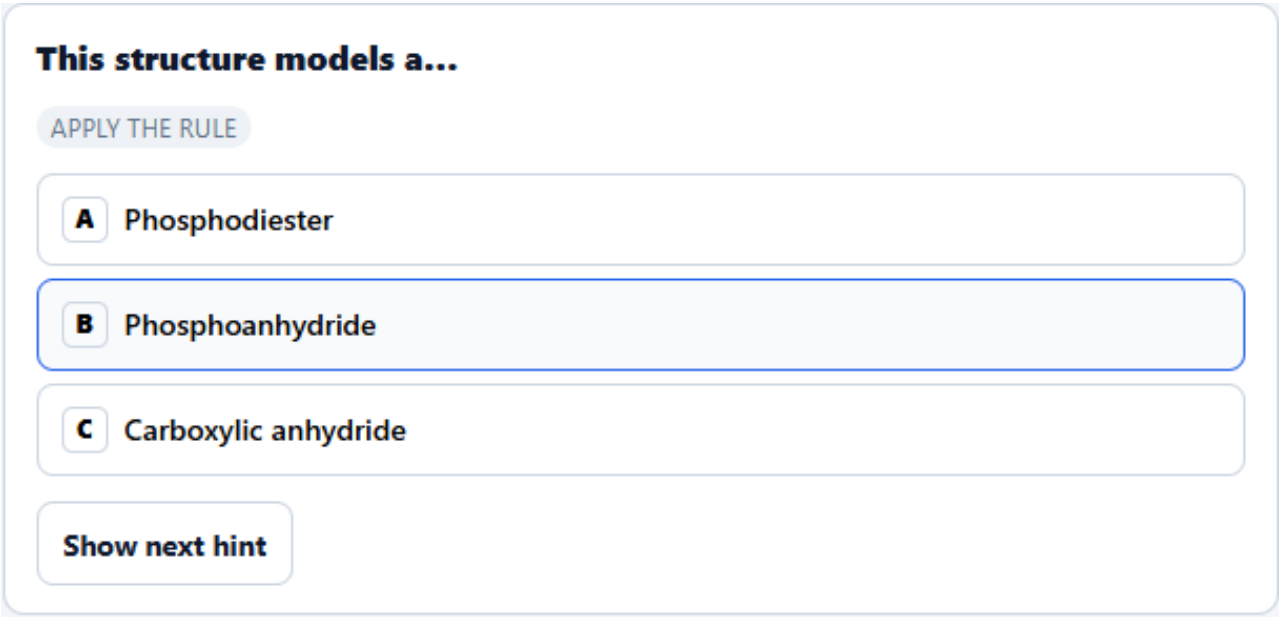
Option 1 (B): Phosphoanhydride

Option 2 (C): Carboxylic anhydride

Screenshot: phosphate-links-stage-2-molecule



Screenshot: phosphate-links-stage-2-question



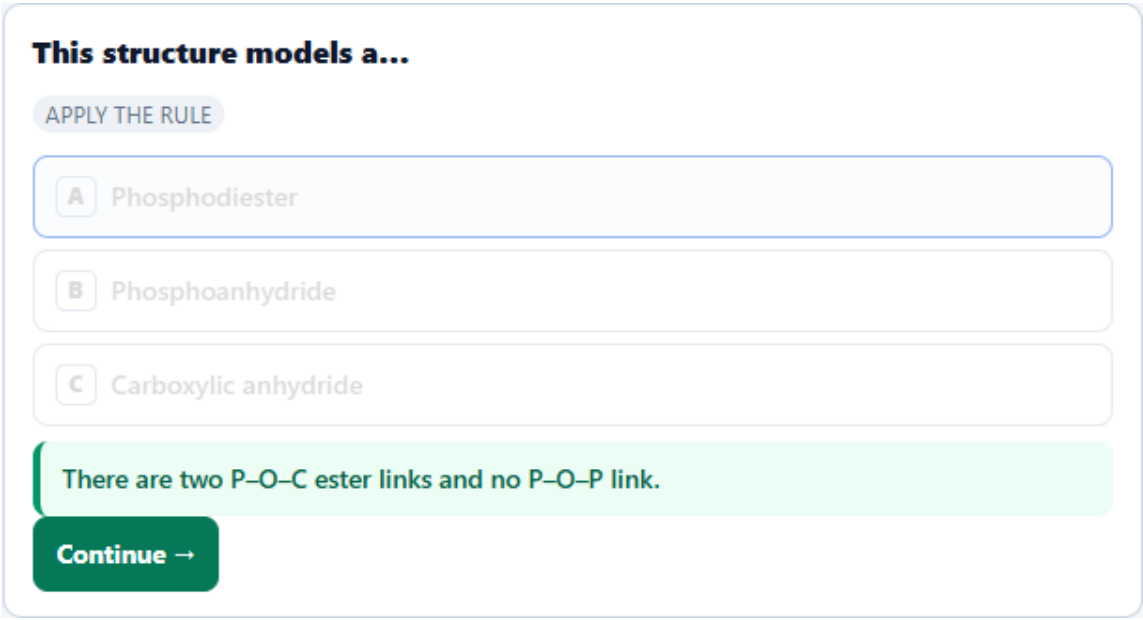
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count oxygen links to carbon groups.

Hint 2: Only one phosphorus is present.

Explanation: There are two P–O–C ester links and no P–O–P link.

Screenshot: phosphate-links-stage-2-feedback



phosphate-links:stage-3 | Why can ATP hydrolysis be favorable overall?

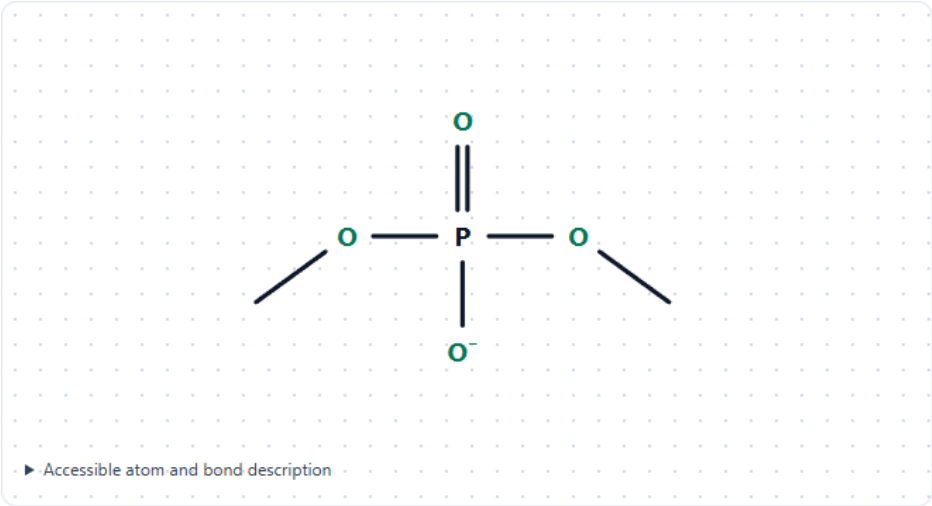
Answer type: choice

Option 0 (A): Products have lower free energy under the stated conditions

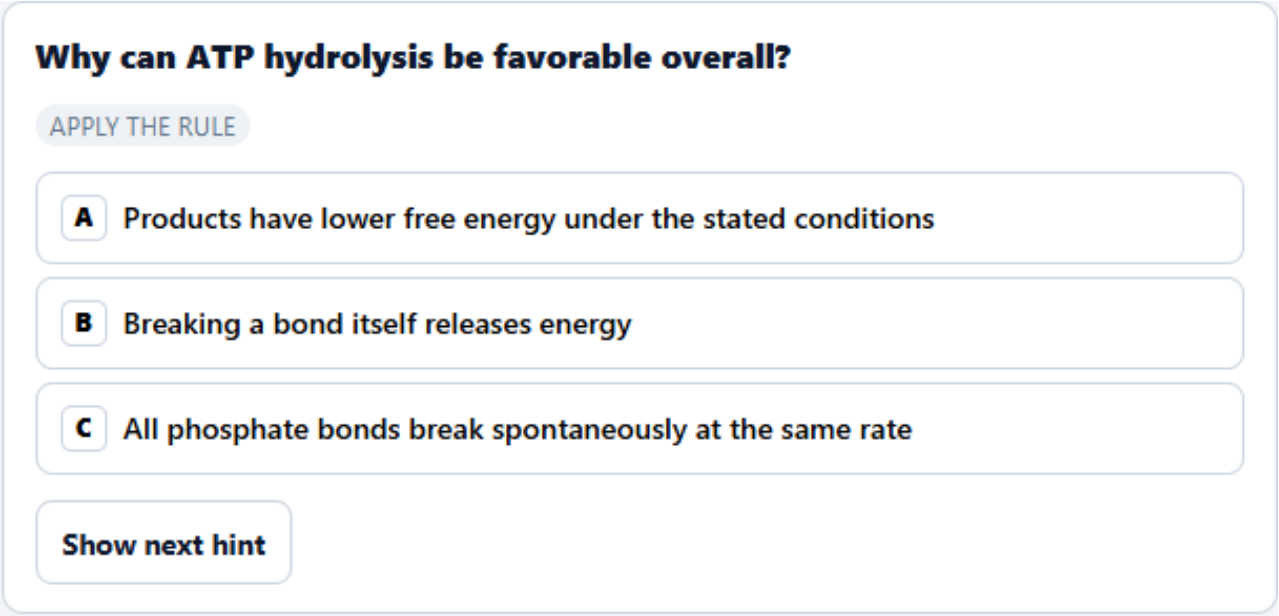
Option 1 (B): Breaking a bond itself releases energy

Option 2 (C): All phosphate bonds break spontaneously at the same rate

Screenshot: phosphate-links-stage-3-molecule



Screenshot: phosphate-links-stage-3-question



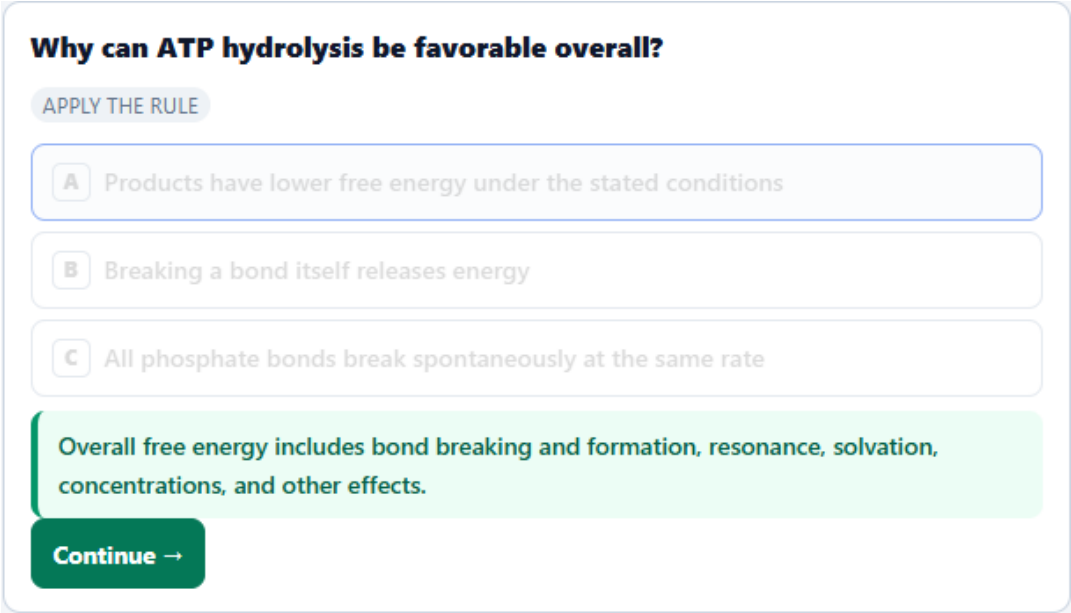
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish a single bond-breaking step from the full reaction.

Hint 2: Thermodynamics does not determine the rate by itself.

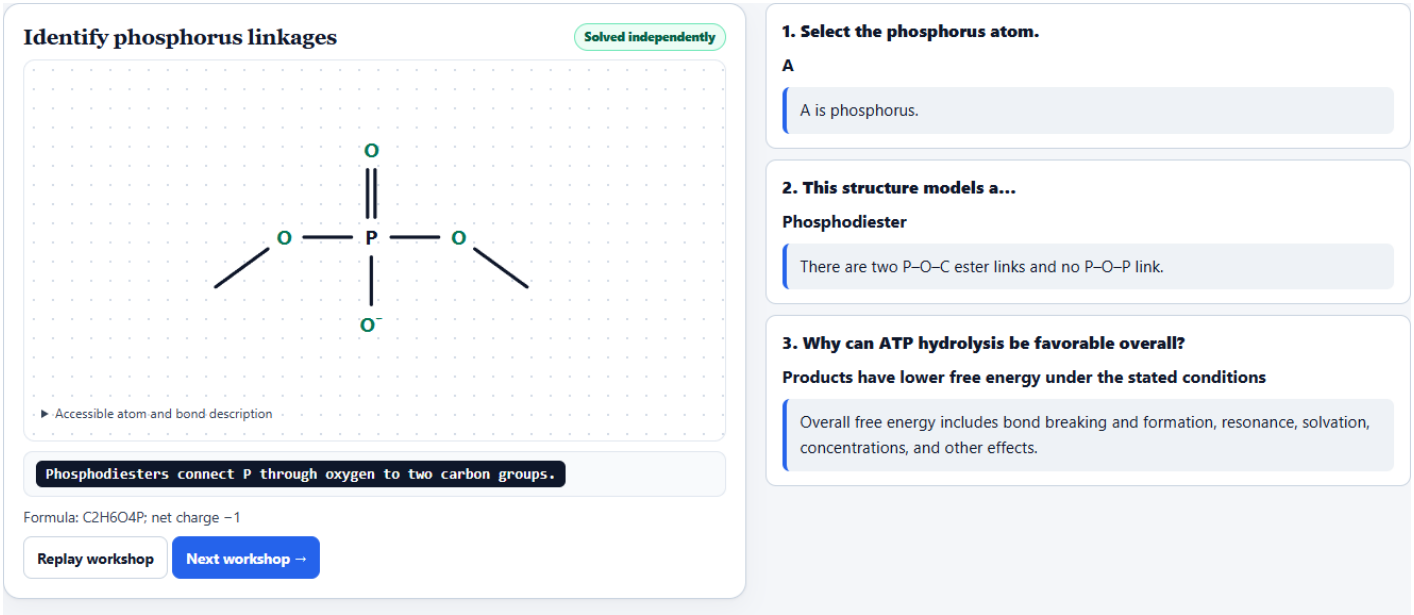
Explanation: Overall free energy includes bond breaking and formation, resonance, solvation, concentrations, and other effects.

Screenshot: phosphate-links-stage-3-feedback



Completion summary: Phosphodiesters connect P through oxygen to two carbon groups.

Screenshot: phosphate-links-complete



Molecular repository: author claims to verify

alanine

Author claims: alanine zwitterion (stereochemistry unspecified) | formula C3H7NO2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=45.00, y=115.00; charge=1; explicit H override=none

1 (B): C; x=105.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=165.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=165.00, y=175.00; charge=0; explicit H override=none

4 (E): O; x=225.00, y=85.00; charge=-1; explicit H override=none

5 (F): C; x=105.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [1, 5, 1]]

glycine

Author claims: glycine zwitterion | formula C2H5NO2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=45.00, y=115.00; charge=1; explicit H override=none

1 (B): C; x=105.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=165.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=165.00, y=175.00; charge=0; explicit H override=none

4 (E): O; x=225.00, y=85.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1]]

glycylglycine

Author claims: glycylglycine zwitterion | formula C4H8N2O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=35.00, y=110.00; charge=1; explicit H override=none

1 (B): C; x=90.00, y=80.00; charge=0; explicit H override=none

2 (C): C; x=145.00, y=110.00; charge=0; explicit H override=none

3 (D): O; x=145.00, y=170.00; charge=0; explicit H override=none

4 (E): N; x=200.00, y=80.00; charge=0; explicit H override=none

5 (F): C; x=255.00, y=110.00; charge=0; explicit H override=none

6 (G): C; x=310.00, y=80.00; charge=0; explicit H override=none

7 (H): O; x=310.00, y=20.00; charge=0; explicit H override=none

8 (I): O; x=365.00, y=110.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 2], [6, 8, 1]]

aminonitrile

Author claims: 2-aminopropanenitrile (stereochemistry unspecified) | formula C3H6N2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=110.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=170.00, y=120.00; charge=0; explicit H override=none

3 (D): N; x=230.00, y=155.00; charge=0; explicit H override=none

4 (E): N; x=110.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 3], [1, 4, 1]]

dimethylphosphate

Author claims: dimethyl phosphate monoanion (formal Lewis representation) | formula C2H6O4P | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): P; x=160.00, y=100.00; charge=0; explicit H override=none

1 (B): O; x=160.00, y=30.00; charge=0; explicit H override=none

2 (C): O; x=90.00, y=100.00; charge=0; explicit H override=none

3 (D): O; x=230.00, y=100.00; charge=0; explicit H override=none

4 (E): O; x=160.00, y=170.00; charge=-1; explicit H override=none

5 (F): C; x=35.00, y=140.00; charge=0; explicit H override=none

6 (G): C; x=285.00, y=140.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [0, 2, 1], [0, 3, 1], [0, 4, 1], [2, 5, 1], [3, 6, 1]]

phosphoricacid

Author claims: phosphoric acid (formal Lewis representation) | formula H3O4P | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): P; x=150.00, y=100.00; charge=0; explicit H override=none

1 (B): O; x=150.00, y=30.00; charge=0; explicit H override=none

2 (C): O; x=80.00, y=100.00; charge=0; explicit H override=none

3 (D): O; x=220.00, y=100.00; charge=0; explicit H override=none

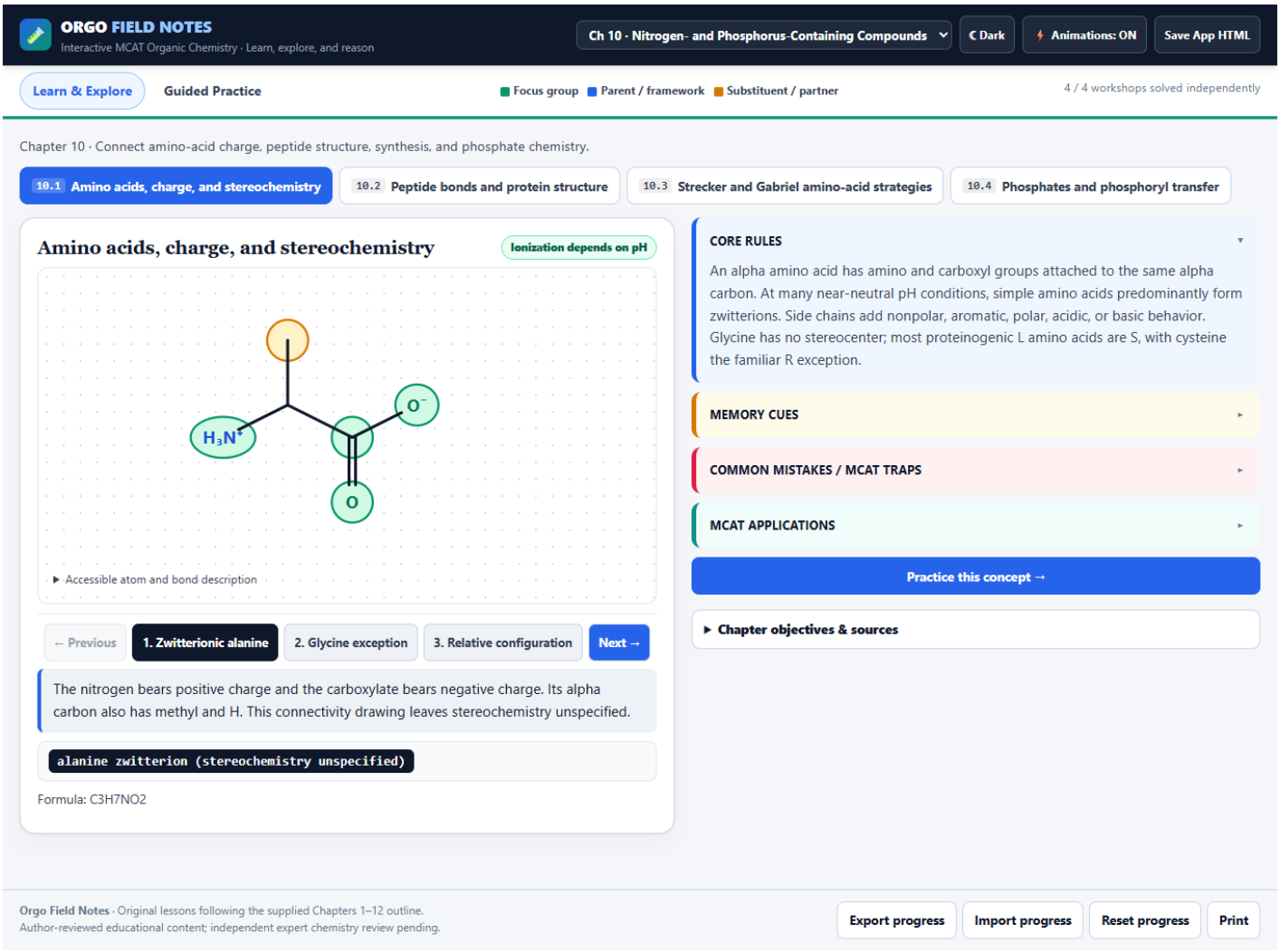
4 (E): O; x=150.00, y=170.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [0, 2, 1], [0, 3, 1], [0, 4, 1]]

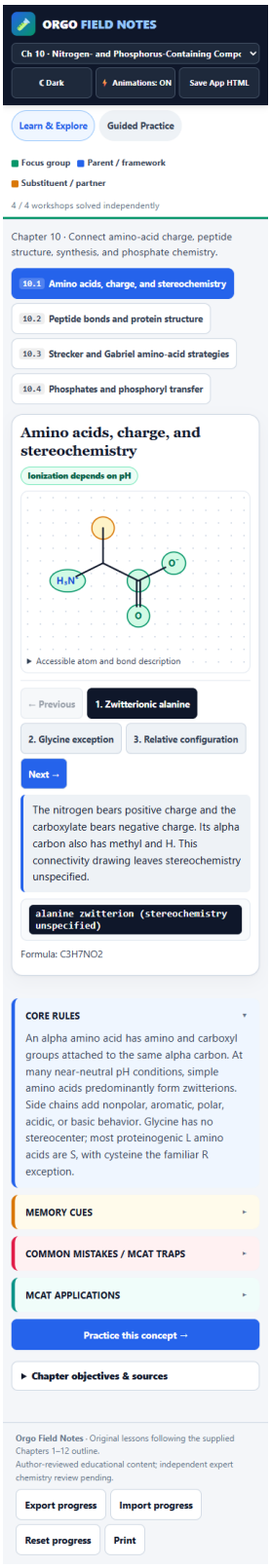
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

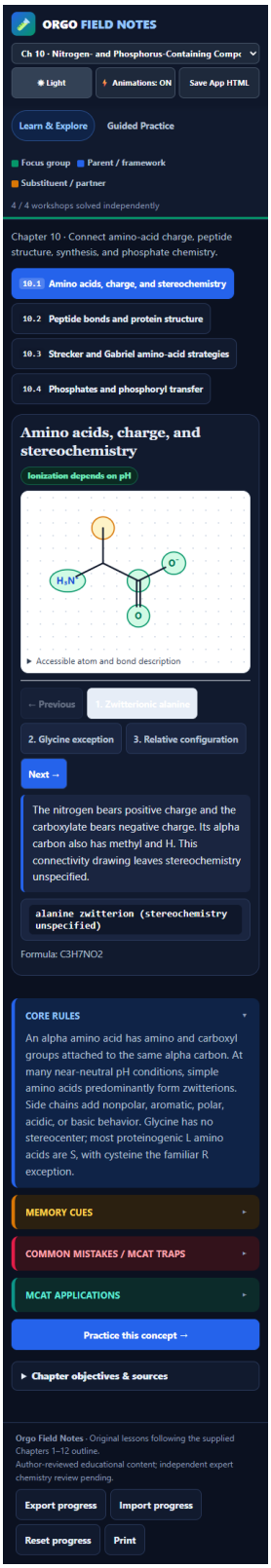
Screenshot: layout-desktop-light



Screenshot: layout-mobile-light



Screenshot: layout-mobile-dark



Execution evidence

55 screenshots and 13 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.